
Supplementary information

**Trends in the seasonal amplitude of
atmospheric methane**

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Supplementary information for

Trends in the seasonal amplitude of atmospheric methane

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References

1. Gaps in NOAA observation data

In this Supplementary section, we provide a comprehensive evaluation and further discussion of the gaps in NOAA observation data and their impacts on long-term trend detection of seasonal cycle amplitude.

We first provided the number of observations for each site from 1984 to 2020 (without missing data, there are 46 8-day flasks in a year) to show the spatial or temporal gaps in the observations (Supplementary Fig. 28). Some sites have started observations from 1994 (e.g., WIS), while some ceased observations after 2010 (e.g. STM). In our study, for the analysis of long-term trends in seasonal CH₄ amplitude, we selected data with each year having at least 23 8-day flasks (the grey squares in Supplementary Fig. 28 represent the gap years for the SCA analysis).

The observations at MLO are complete between 1984 and 2020, and show a significant increasing trend of CH₄ SCA. To test the impacts of temporal gaps on the long-term trend detection, we added different numbers of artificial gap years at MLO. First, for each number of artificial gap years (from 1 year to 30 years), we have applied Monte Carlo method to take 1000 time series with randomly removing artificial gap years at MLO, and calculated the trends in SCA of these artificial time series. The proportion of the artificial time series with significant SCA trend ($p < 0.05$) decreases with the increasing in artificial gap years (blue line in Supplementary Fig. 29). When the artificial temporal gaps are less than or equal to 7 years, over 95% of the time series show a significant SCA trend. Thus, the sites with temporal gap years ≤ 7 years could have reliable trends. For the latitudinal gradient of SCA trend analysis, if the 8 sites with more than 7 gap years are excluded, the SCA trend at the other 10 sites (≤ 7 years) still significantly decreases with latitude ($-1.0 [-1.5 - -0.5]$ ppb decade⁻¹ per 10 degrees of latitude, $p < 0.01$), compared with the result of all 18 sites ($-0.8 [-1.3 - -0.4]$ ppb decade⁻¹ per 10 degrees of latitude, $p < 0.01$). For the spatial gaps, we find that at least one site for every 10 degrees of latitude north of 10°N, well representing their latitudinal bands. There are seven sites in the Southern Hemisphere (Supplementary Fig. 2), and showing consistent increasing trends

as the long-term observations at high quality site SPO. Thus, these gaps in the observational data have limited impacts on the findings in our study.

2. Uncertainties of wetland emissions

For evaluating the uncertainties of wetland emissions, we have compiled 31 datasets of wetland emissions, including modelling results of ORCHIDEE, LPJ-EOSIM¹ and from Global Carbon Project^{2,3} (GCP-CH₄), WETCHARTS⁴, and atmospheric inversions from CAMS⁵, as well as upscaling wetland methane emissions (UpCH₄)⁶ and Boreal-Arctic wetland emissions from the causality-guided machine learning model⁷.

In the northern high latitudes, 29 out of 31 datasets indicate an increasing trend of wetland emissions (median: 0.3 Tg CH₄ yr⁻², ranging from -0.3 to 1.6 Tg CH₄ yr⁻²; Supplementary Fig. 30), supporting the increased wetland emissions in the northern high latitudes inferred from the decreasing trend of CH₄ SCA in this study. Moreover, 28 datasets show increase in tropical wetland emissions after 2000 (median: 2.5 Tg CH₄ yr⁻², ranging from -0.9 to 11.3 Tg CH₄ yr⁻²). The increase in emissions (amplitude) from northern and southern tropical wetlands could lower atmospheric CH₄ SCA, which contradicts to the observed increasing trend of SCA in the tropics (Fig. 1). The increasing trends of wetland emissions in northern high latitudes (0.9 Tg CH₄ yr⁻²) and tropics (2.6 Tg CH₄ yr⁻²) from ORCHIDEE are close to ensemble mean increase rate of 0.6 Tg CH₄ yr⁻² and 3.3 Tg CH₄ yr⁻² from northern high wetland and tropical wetland (Supplementary Fig. 30), respectively.

3. Uncertainties introduced by offline OH concentration

For evaluating the uncertainties introduced by offline OH concentration, we have quantified the differences between the offline OH concentration forced by the observed CH₄ concentration (OH baseline; Supplementary Fig. 31) and the OH concentration forced by the CH₄ concentration in our T1 experiments simulated with OH concentrations from INCA (T1-INCA) and GEOS-Chem (T1-GEOSChem), respectively. We found that both the OH concentrations forced by CH₄ from T1-INCA and T1-GEOSChem are close to the OH baseline

(Supplementary Fig. 31), with discrepancies lower than 0.7% in these two periods 1984-1989 (-0.3% for T1-INCA and 0.1% for T1-GEOSChem) and 2015-2020 (-0.1% for T1-INCA and 0.7% for T1-GEOSChem). Meanwhile, the discrepancies of 0.7% in OH concentrations would lead to an error of ~1.3 ppb (0.7% multiple the S_{OH}) in the simulation of atmospheric CH_4 concentration. This indicates that the uncertainty introduced by the offline and online OH has a minimal impact on the results.

4. Seasonal cycle amplitude of other trace gases and MALTA model

We have collected available data for 42 compounds (Table S3), with a span of lifetimes, from NOAA' Long-term Observations of Greenhouse gases and Ozone-depleting Substances (LOGOS; <https://gml.noaa.gov/hats/>) and from the Advanced Global Atmospheric Gases Experiment (AGAGE; <https://www-air.larc.nasa.gov/missions/agage/data>). Several compounds have been ruled out following three criteria as below.

1) For evaluating long-term trends in SCA, 23 out of 42 compounds with continuous observations less than 20 years are excluded. Although the observations of HFC-125 and HCFC-124 from AGAGE started since 1998, they have discontinued measurements due to the replacement of instruments in 2000s. These two compounds were excluded to be on the safe side.

2) Then, we excluded the compounds, whose sink is not dominated by tropospheric OH concentration. Specifically, the stratospheric photolysis is the primary sink of CFC-11, CFC-12, CFC-113, CCl_4 , halon-1211, halon-1301 and halon-2402 (see Fig 3.5 in Ko et al. (ref.⁸)). The lifetime of C_2Cl_4 is strongly affected by the tropospheric Cl sink (101 days without Cl sink; 68 days with Cl sink)⁹. The ocean and soil uptake contributes to over 50% of total sink of CH_3Br ^{10,11}, while multiple lines of evidence support uptake by plants as the dominant removal mechanism of atmospheric COS^{12,13}.

3) Finally, to mitigate the impact of emissions on the changes of SCA, the compounds with strong seasonal natural or anthropogenic emissions are also excluded. There is a well-documented seasonal variation in global anthropogenic emissions of HCFC-22 and HFC-134a, with the summer emissions of these refrigerants being two to three times higher than in winter¹⁴. Furthermore, the seasonally varying anthropogenic emission has been shown to contribute to the seasonal cycle of atmospheric HCFC-152a¹⁵. The global emissions of CHCl_3 are mainly attributed to natural emissions¹⁶ (90.5%), including oceanic flux, soil and termites. Similarly, the global emissions of CH_3Cl are presumably caused by large tropical terrestrial sources, while anthropogenic emissions only contributed to 5.4% of global emissions¹⁷. The inversion results suggest significant seasonal variations in the natural CH_3Cl sources¹⁷.

As a result, four compounds (CH_2Cl_2 , CH_3CCl_3 , HCFC-141b and HCFC-142b) are applied for the SCA analysis. The seasonal cycle of CH_2Cl_2 is mainly due to its reaction with the OH radical, with maxima in late winter/early spring and corresponding minima in late summer or early autumn¹⁸. With regard to the emissions of CH_2Cl_2 , it has a range of industrial applications and roughly 90% of total emissions have been estimated to be anthropogenic¹⁹, and no significant seasonal variation in industrial emissions has been reported²⁰. The emissions of CH_3CCl_3 have declined substantially due to the revised and amended Montreal Protocol on Substances that Deplete the Ozone Layer¹⁹, suggesting that the seasonal variations of CH_3CCl_3 are dominated by the OH sink. Given the lack of available references/datasets showing the seasonality in HCFC-141b and HCFC-142b emissions, these two compounds are also included in our analysis.

The changes in OH sink of each compound, contributed by both its atmospheric concentration and OH concentration, may potentially play a key role on the SCA. To quantify the contributions of OH concentration to the observed SCA, we used a zonally averaged global atmospheric transport model MALTA²¹ to attribute the changes in SCA of these four compounds. The horizontal resolution of MALTA is every 10° in latitude and the bounds of altitude are at 30 levels (29 layers) from 1000 hPa to 10hPa. For each compound, the MALTA is forced by dynamic transport and meteorological fields (MERRA2; ref. ²¹) with varying OH concentration from GEOS-Chem (M1 simulation). The emissions are assumed to be distributed

latitudinally by Gross Domestic Product (GDP) in a given year, and without seasonal variations. The year-to-year emissions are iteratively adjusted by gradient descent to ensure the annual atmospheric concentration from simulation are consistent with that from observations at MLO (Eq. S1).

$$E_{n+1,y} = E_{n,y} - (C_{n,y} - C_{MLO}) * cf * g \quad \text{Eq. (S1)}$$

Where C_{MLO} is the observed atmospheric concentration at MLO. $E_{n,y}$ and $C_{n,y}$ is the global emission and simulated 10-30°N average atmospheric concentration in the third layers (~2900m) for each year y , in the n th iteration. The cf is the conversion factor from emissions to concentrations and g is a constant factor of decreasing rate. We performed 5 years spin-up for short-lived gas (CH_2Cl_2) and 20 years spin-up for the other three gases. We also performed M2 simulation with fixed OH concentration, while emissions and atmospheric concentration involved in the oxidation reaction for each compound are kept the same as M1 simulation. Then we applied SSA frequency decomposition to extract the seasonal cycle of atmospheric concentrations of these four compounds from observations and simulations. The first and last year of the SCA are excluded in the trend analysis.

Without seasonal variations in emissions, the seasonal cycle of CH_2Cl_2 and CH_3CCl_3 from our simulations show a good match to the SCA from observations (Supplementary Fig. 32). However, the simulations of HCFC-141b and HCFC-142b fail to replicate the observed seasonal variations in atmospheric concentration. The simulated seasonal atmospheric concentration of HCFC-141b and HCFC-142b peaks in May and then decreases, driven by an increase in the atmospheric sink of OH radicals, while the seasonal concentration from observations increases in summer (June, July and August). The increase in the seasonal concentration is likely to be contributed by the seasonality in emissions, with higher emissions in summer as components of some refrigerant blends. Thus, we deduce that the SCA of HCFC-141b and HCFC-142b are dominated by the emissions, rather than the OH sink.

The seasonal cycles of CH_2Cl_2 and CH_3CCl_3 are dominated by the OH sink. The M1 simulation could capture the seasonal cycle of these two compounds and the changes of SCA over the past

two decades (Supplementary Fig. 33). The M2 simulation with fixed OH concentration show a worse performance on capturing the trends in SCA than M1 simulation (Supplementary Fig. 33), indicating OH concentration has contributed to the trends in SCA of these two compounds. The increasing trend of CH_2Cl_2 SCA from our simulations ($0.563 \text{ ppt yr}^{-1}$) is contributed by both rising CH_2Cl_2 concentration ($0.430 \text{ ppt yr}^{-1}$; 76%) and increasing OH concentration ($0.123 \text{ ppt yr}^{-1}$; 24%). The decreasing trend of $\log(\text{CH}_3\text{CCl}_3 \text{ SCA})$ ($-0.1768 \text{ ppt yr}^{-1}$) is dominated by the decline of CH_3CCl_3 atmospheric concentration ($-0.1770 \text{ ppt yr}^{-1}$; 101%), from 120.7 ppt in 1994 to 1.4 ppt in 2020, partly offset by the enhanced OH concentration ($0.002 \text{ ppt yr}^{-1}$; 1%).

Overall, the coincident evidences from CH_4 SCA and CH_2Cl_2 SCA suggest an increase in the tropospheric OH concentration. The positive correlations between the interannual variabilities of CH_4 amplitude and CH_2Cl_2 amplitude at MLO ($r = 0.50 - 0.61$, $p = 0.00 - 0.02$) and KUM ($r = 0.39 - 0.49$, $p = 0.05 - 0.11$; Supplementary Fig. 27) further support that the increasing trends in SCA of these gases are likely both driven by changes in OH sink, especially the OH concentration.

It is worth noting that the simple atmospheric transport model MALTA used here only provides a perspective on the contributions of OH concentration to SCA, considering MALTA's horizontal resolution is 10° in latitude. Moreover, the latitudinal distributions and temporal trends of emissions have large uncertainties^{22,23}. The seasonal cycle of trace gases provides a valuable opportunity for better understanding the trends in OH concentration, but further research is needed to reduce the uncertainties (e.g., emission inventories with spatial distribution, seasonality of emissions etc.).

Supplementary Table 1

List of the CH₄ measurement sites used (Lan et al. (ref.²⁴)).

Station Name	Station Code	Latitude	Longitude	Start Year	Number of years
Alert, Nunavut, Canada	ALT	82.5	-62.5	1985	35
Ny-Alesund, Svalbard, Norway and Sweden	ZEP	78.9	11.9	1994	27
Barrow Atmospheric Baseline Observatory, United States	BRW	71.3	-156.6	1983	38
Ocean Station M, Norway	STM	66.0	2.0	1983	27
Storhofdi, Vestmannaeyjar, Iceland	ICE	63.4	-20.3	1992	25
Cold Bay, Alaska, United States	CBA	55.2	-162.7	1983	35
Mace Head, County Galway, Ireland	MHD	53.3	-9.9	1991	28
Shemya Island, Alaska, United States	SHM	52.7	174.1	1985	33
Niwot Ridge, Colorado, United States	NWR	40.1	-105.6	1983	37
Wendover, Utah, United States	UTA	39.9	-113.7	1993	28
Tae-ahn Peninsula, Republic of Korea	TAP	36.7	126.1	1990	30
Tudor Hill, Bermuda, United Kingdom	BMW	32.3	-64.9	1989	29
Weizmann Institute of Science at the Arava Institute, Ketura, Israel	WIS	30.0	35.1	1995	25
Izana, Tenerife, Canary Islands, Spain	IZO	28.3	-16.5	1991	29
Sand Island, Midway, United States	MID	28.2	-177.4	1985	35
Key Biscayne, Florida, United States	KEY	25.7	-80.2	1983	32
Mauna Loa, Hawaii, United States	MLO	19.5	-155.6	1983	38
Cape Kumukahi, Hawaii, United States	KUM	19.5	-154.8	1983	37
Mariana Islands, Guam	GMI	13.4	144.7	1983	31
Ragged Point, Barbados	RPB	1.2	-59.4	1988	33
Mahe Island, Seychelles	SEY	-4.7	55.5	1983	32
Ascension Island, United Kingdom	ASC	-8.0	-14.4	1983	38
Tutuila, American Samoa	SMO	-14.3	-170.6	1983	37
Cape Grim, Tasmania, Australia	CGO	-40.7	144.7	1984	35
Palmer Station, Antarctica, United States	PSA	-64.8	-64.1	1983	34
Syowa Station, Antarctica, Japan	SYO	-69.0	39.6	1986	26
South Pole, Antarctica, United States	SPO	-90.0	59.0	1983	38

#: Record in each year has at least 23 8-day flasks

Supplementary Table 2

Descriptions and access information of the datasets.

Dataset	Short description	URL
Daily CH ₄ concentration from GESO-Chem simulation	Simulated CH ₄ concentration	https://doi.org/10.6084/m9.figshare.28425077
Atmospheric CH ₄ site data, NOAA	Atmospheric CH ₄ concentration	https://gml.noaa.gov/ccgg/trends_ch4/
Aircraft data, NOAA	Atmospheric CH ₄ concentration	https://gml.noaa.gov/aftp/data/trace_gases/ch4/pfp/aircraft/
EDGAR v7.0	Anthropogenic emissions	https://edgar.jrc.ec.europa.eu/dataset_ghg70
CEDS 2021	Anthropogenic emissions	https://data.pnnl.gov/dataset/CEDS-4-21-21
BB4CMIP	Biomass burning emissions	https://esgf-node.llnl.gov/projects/input4mips/
GFED v4.1s	Biomass burning emissions	https://www.geo.vu.nl/~gwerf/GFED/GFED4/
ORCHIDEE	Wetland emissions	On request: Peng et al. (ref. ²⁵)
LPJ-EOSIM	Wetland emissions	https://doi.org/10.5067/Community/LPJ-EOSIM/LPJ_EOSIM_L2_MCH4E.001
CAMS	Inversion of wetland emissions	https://www.ecmwf.int/en/forecasts/dataset/cams-greenhouse-gas-ghg-flux-inversions
LMDZ-INCA OH concentration	Atmospheric OH concentration	On request: Peng et al. (ref. ²⁵)
GEOS-Chem OH concentration	Atmospheric OH concentration	https://doi.org/10.6084/m9.figshare.28425077
OH fields simulated by ESMs	Atmospheric OH concentration	https://esgf-data.dkrz.de/search/cmip6-dkrz/
MERRA2	Hourly climate forcing data	https://gmao.gsfc.nasa.gov/reanalysis/MERRA-2
CRU TS v4.05	Monthly temperature and precipitation	https://crudata.uea.ac.uk/cru/data/hrg/cru_ts_4.05
MSWEP	Monthly precipitation	http://www.gloh2o.org/mswep
ERA5 reanalysis data	Monthly temperature, precipitation and soil temperature	https://www.ecmwf.int/en/forecasts/dataset/ecmwf-reanalysis-v5
GLEAM v3.8a	Monthly root-zone soil moisture data	https://www.gleam.eu

Supplementary Table 3

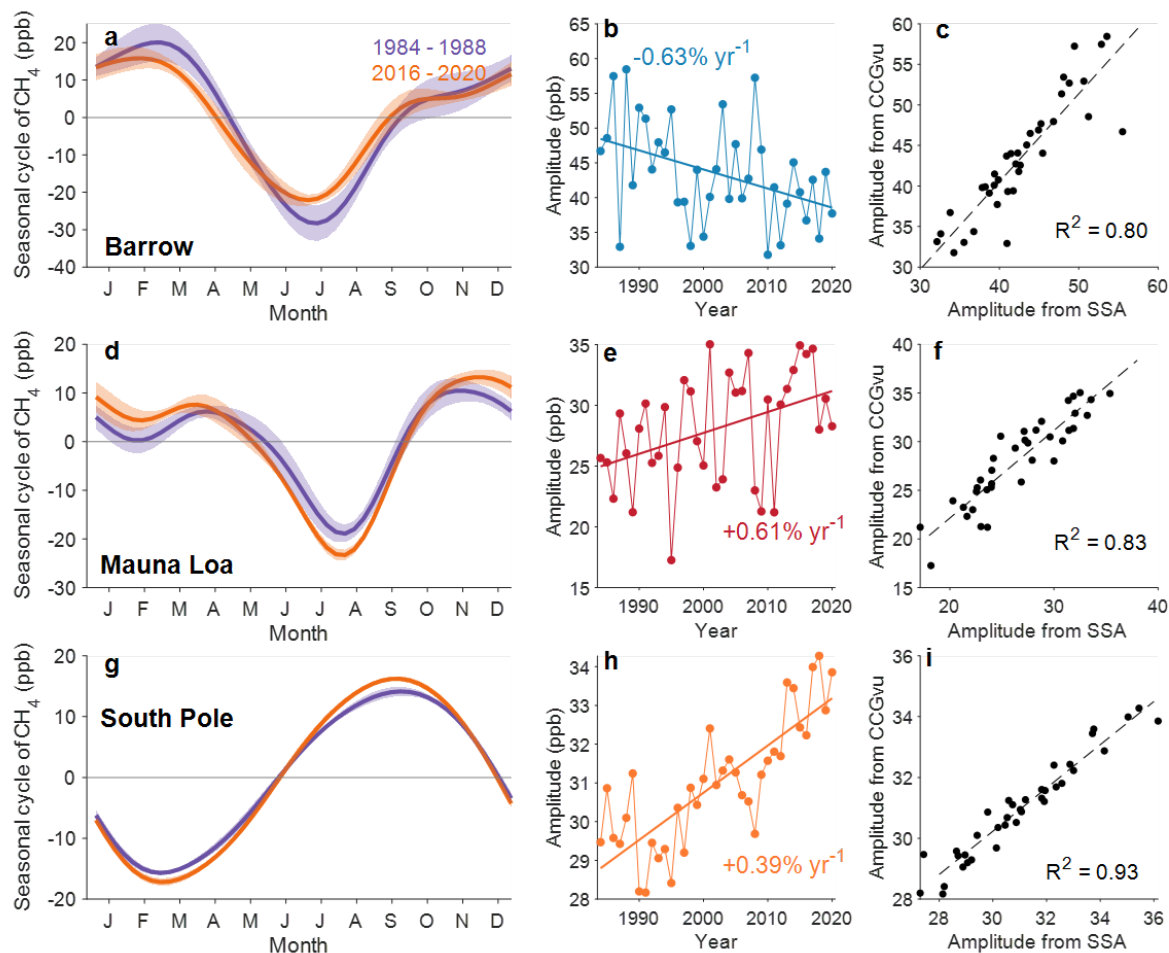
Compounds from NOAA' Long-term Observations of Greenhouse gases and Ozone-depleting Substances (LOGOS; <https://gml.noaa.gov/hats/>) and from the Advanced Global Atmospheric Gases Experiment (AGAGE; <https://www-air.larc.nasa.gov/missions/agage/data>).

The grey shading indicates the compounds with short-term observations less than 20 years. The blue shading indicates the compounds' sink is not dominated by OH, and the orange shading indicates the compounds with strong seasonal natural or anthropogenic emissions.

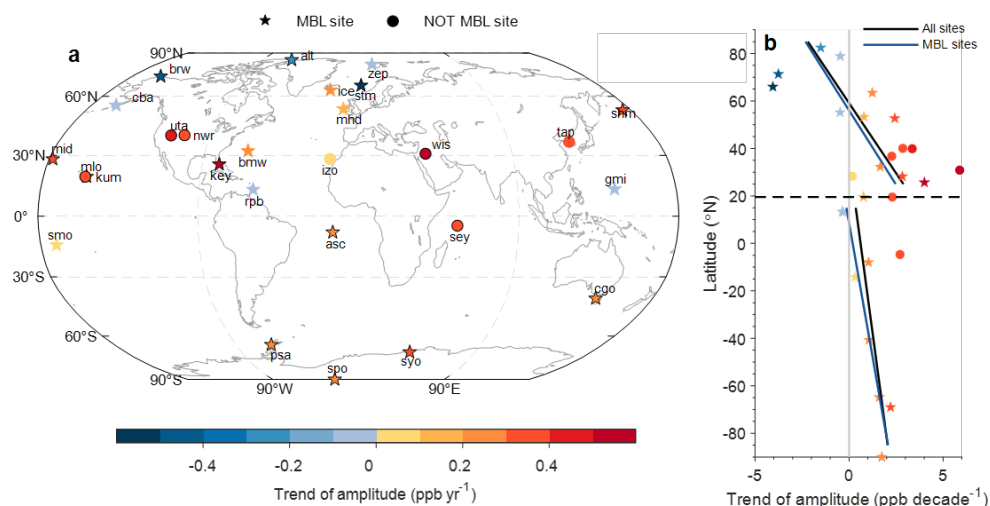
Common Name	Formula	NOAA Start year	AGAGE Start year
Methyl Chloroform	CH ₃ CCl ₃	1994	1979
Dichloromethane	CH ₂ Cl ₂	1995	1995
HCFC-141b	CH ₃ CCl ₂ F	1994	1995
HCFC-142b	CH ₃ CClF ₂	1994	1995
CFC-114	CClF ₂ CClF ₂	\	2004
CFC-115	CClF ₂ CF ₃	\	2004
CFC-13	CClF ₃	\	2004
HFC-23	CHF ₃	\	2008
HFC-32	CH ₂ F ₂	2009	2004
HFC-125	CHF ₂ CF ₃	2007	1998*
HFC-143a	C ₂ H ₃ F ₃	2007	2004
HFC-227ea	CF ₃ CH ₂ CF ₃	2011	2005
HFC-365mfc	CF ₃ CH ₂ CF ₂ CH ₃	2009	2003
HFC-236fa	CF ₃ CH ₂ CF ₃	2015	2007
HFC-245fa	CHF ₂ CH ₂ CF ₃	\	2005
HFC-43-10mee	C ₅ H ₂ F ₁₀	\	2011
HCFC-124	CHClF ₂ CF ₃	\	1998*
HCFC-132b	C ₂ H ₂ Cl ₂ F ₂	\	2013
HCFC-133a	CF ₃ CH ₂ Cl	\	2013
PFC-116	C ₂ F ₆	2015	2004
PFC-14	CF ₄	2015	2004
PFC-218	C ₃ F ₈	\	2004
PFC-318	c-C ₄ F ₈	\	2010
Nitrogen Trifluoride	NF ₃	2015	2014
Sulfuryl Fluoride	SO ₂ F ₂	2015	2005
Ethane	C ₂ H ₆	2014	\
Propane	C ₃ H ₈	2014	\
CFC-11	CCl ₃ F	1980	1979
CFC-12	CCl ₂ F ₂	1980	1979
CFC-113	CCl ₂ FCClF ₂	1995	1983
Carbon Tetrachloride	CCl ₄	1990	1979
Perchloroethylene	C ₂ Cl ₄	1994	2000
halon-1211	CBrClF ₂	1994	1995
halon-1301	CBrF ₃	2005	1998
halon-2402	C ₂ Br ₂ F ₄	1995	2004
Methyl Bromide	CH ₃ Br	1996	1998
Carbonyl Sulfide	COS	2000	\
HCFC-22	CHClF ₂	1994	1998
HFC-134a	CH ₂ FCF ₃	1995	1995
HFC-152a	CH ₃ CHF ₂	2000	1995
Chloroform	CHCl ₃	\	1994
Methyl Chloride	CH ₃ Cl	1996	1994

*: discontinued measurements as a result of the replacement of instruments in 2000s

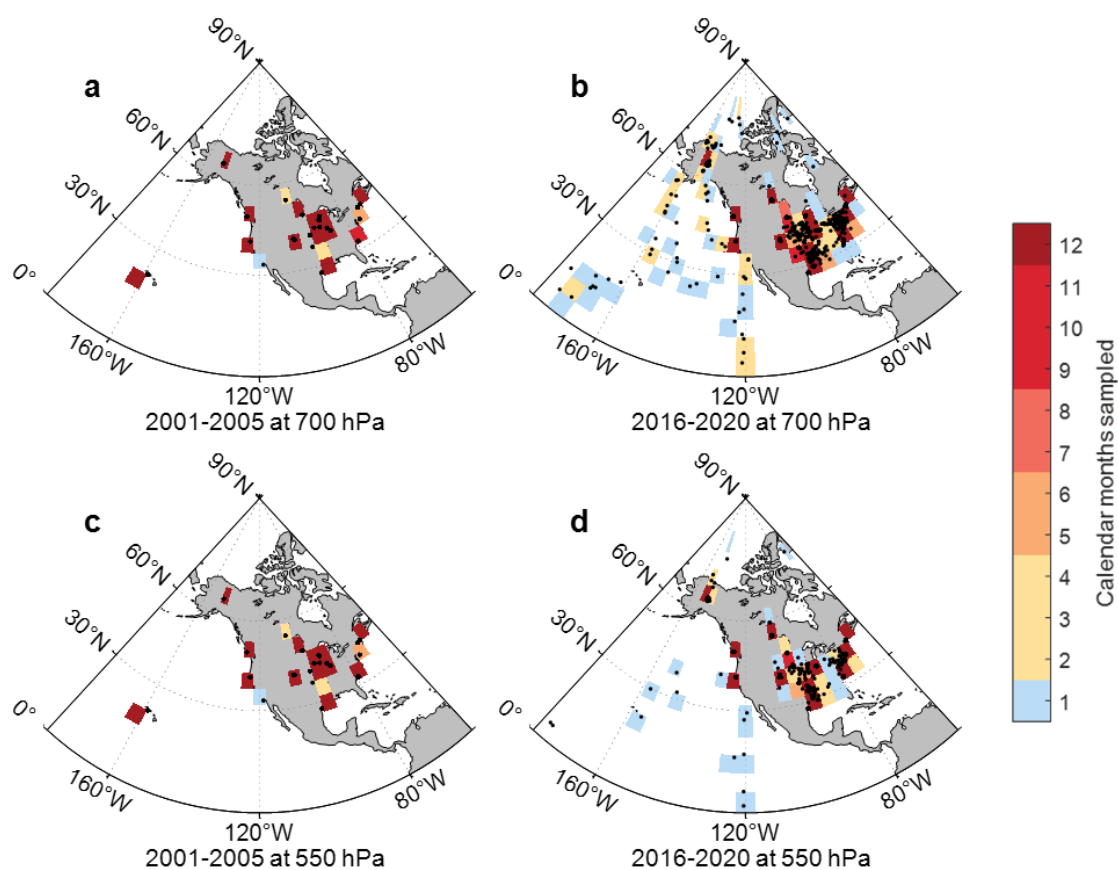
Supplementary Figure 1. Long-term changes in seasonal CH₄ cycle derived from CCGCRV at Barrow, Mauna Loa and South Pole. Seasonal CH₄ cycles derived by CCGCRV²⁶ from the observations at Barrow (**a**), Mauna Loa (**d**), and South Pole (**g**) for the periods 1984-1988 and 2016-2020 and their peak-to-trough amplitude (**b**, **e**, **h**). The comparison between the seasonal amplitude derived from SSA and CCGCRV at Barrow (**c**), Mauna Loa (**f**), and South Pole (**i**).



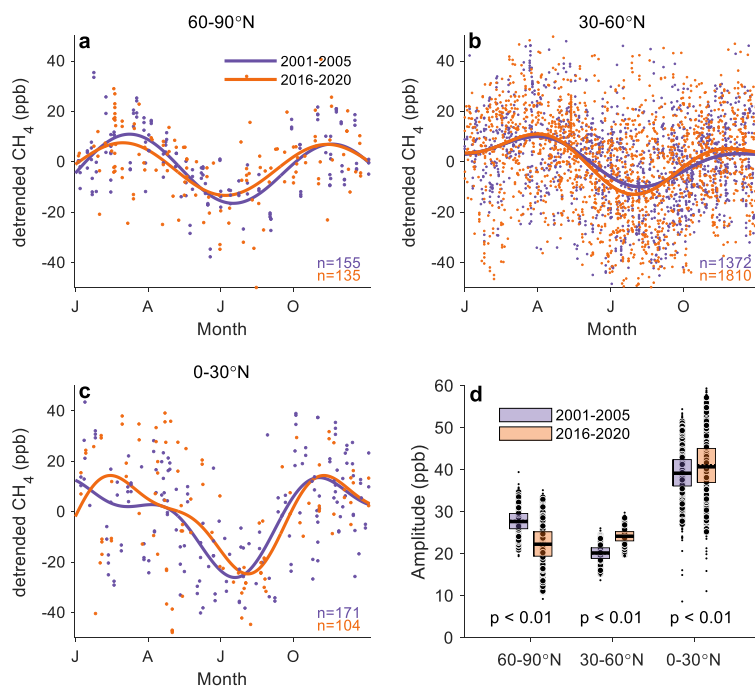
Supplementary Figure 2. Long-term trends of site-level CH₄ amplitude from observations. The trends of seasonal CH₄ amplitude at all sites (a) and the latitudinal gradient of SCA at the sites north of MLO (19.5°N–90.0°N) and south of MLO (90°S–19.5°N) (b). The slope of the signal changes most abruptly at approximately 19.5°N, which is detected by the MATLAB program ‘findchangepts’.



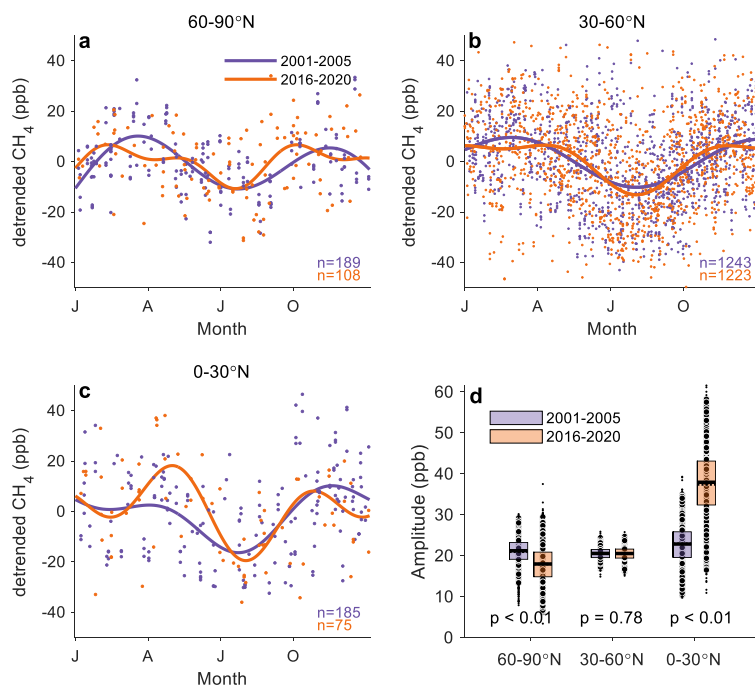
Supplementary Figure 3. Sampling locations of NOAA aircraft observations. Sampling locations at 2500m-3500m during 2001-2005 (a) and 2016-2020 (b) and at 4500m-5500m during 2001-2005 (c) and 2016-2020 (d). Colours indicate the number of calendar months sampled in 5°x5° bins.



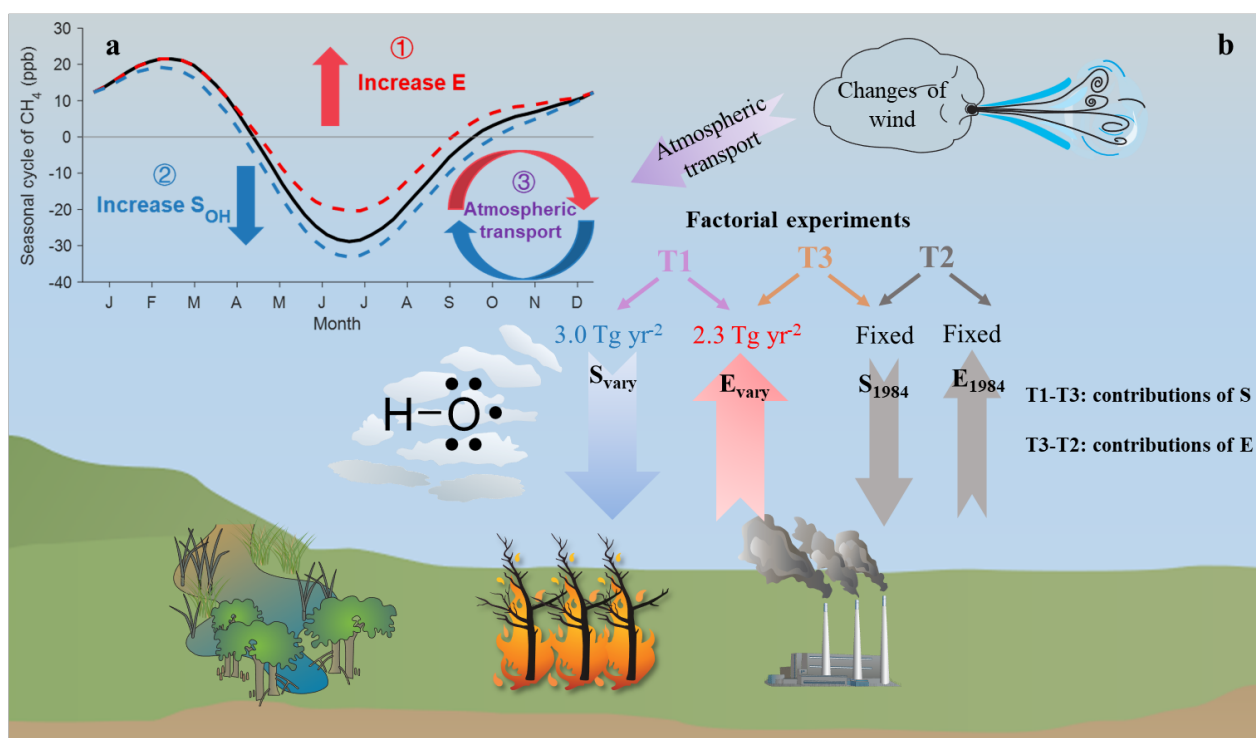
Supplementary Figure 4. Seasonal CH₄ cycle observed by aircraft in 2001 to 2005 and 2016 to 2020. Seasonal cycles of CH₄ at 2500m-3500m for three latitudinal bands in both periods. Curves show the seasonal cycles fit to the data. The distribution of the seasonal amplitude for randomly selecting 90% samples by 1000 times in the periods of 2001-2005 and 2016-2020. The p-value determined by the t-test represents the significance of the difference in amplitude distributions between the two periods.



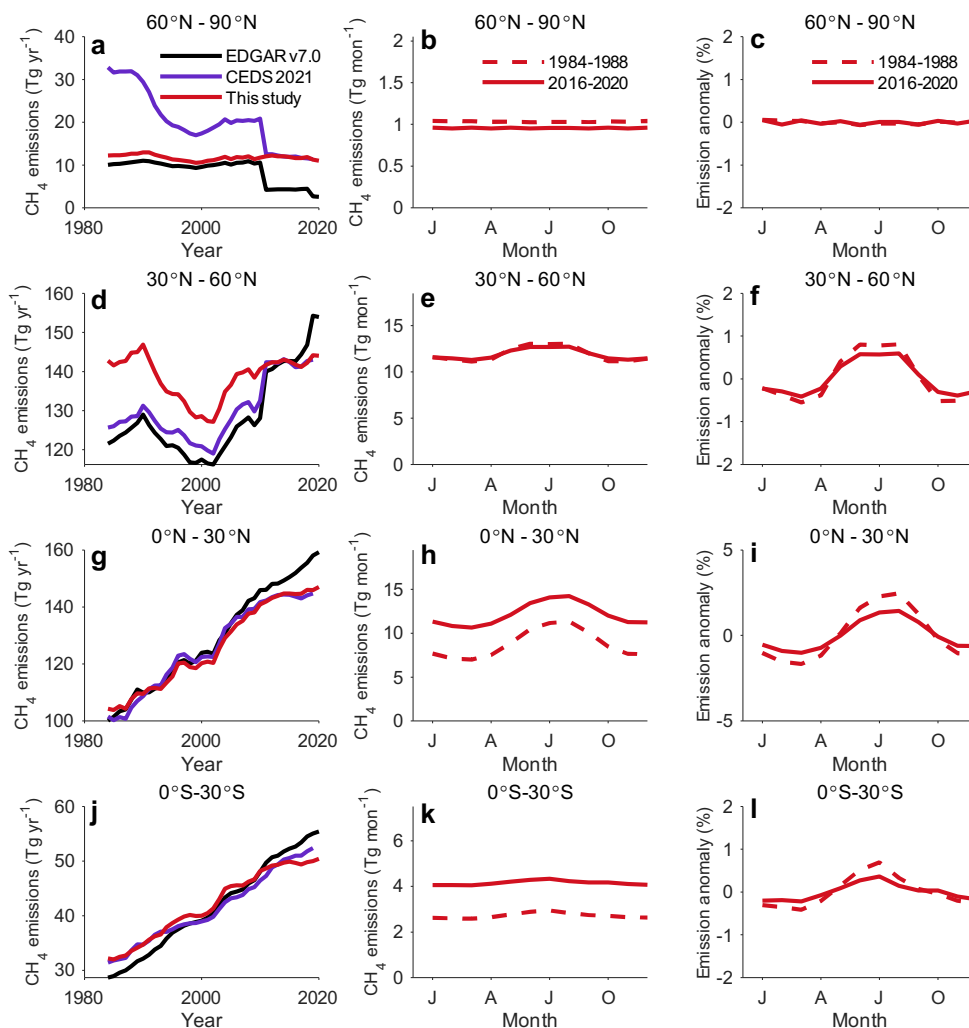
Supplementary Figure 5. Seasonal CH₄ cycle observed by aircraft in 2001 to 2005 and 2016 to 2020. Seasonal cycles of CH₄ at 4500m-5500m for three latitudinal bands in both periods. Curves show the seasonal cycles fit to the data. The distribution of the seasonal amplitude for randomly selecting 90% samples by 1000 times in the periods of 2001-2005 and 2016-2020. The p-value determined by the t-test represents the significance of the difference in amplitude distributions between the two periods.



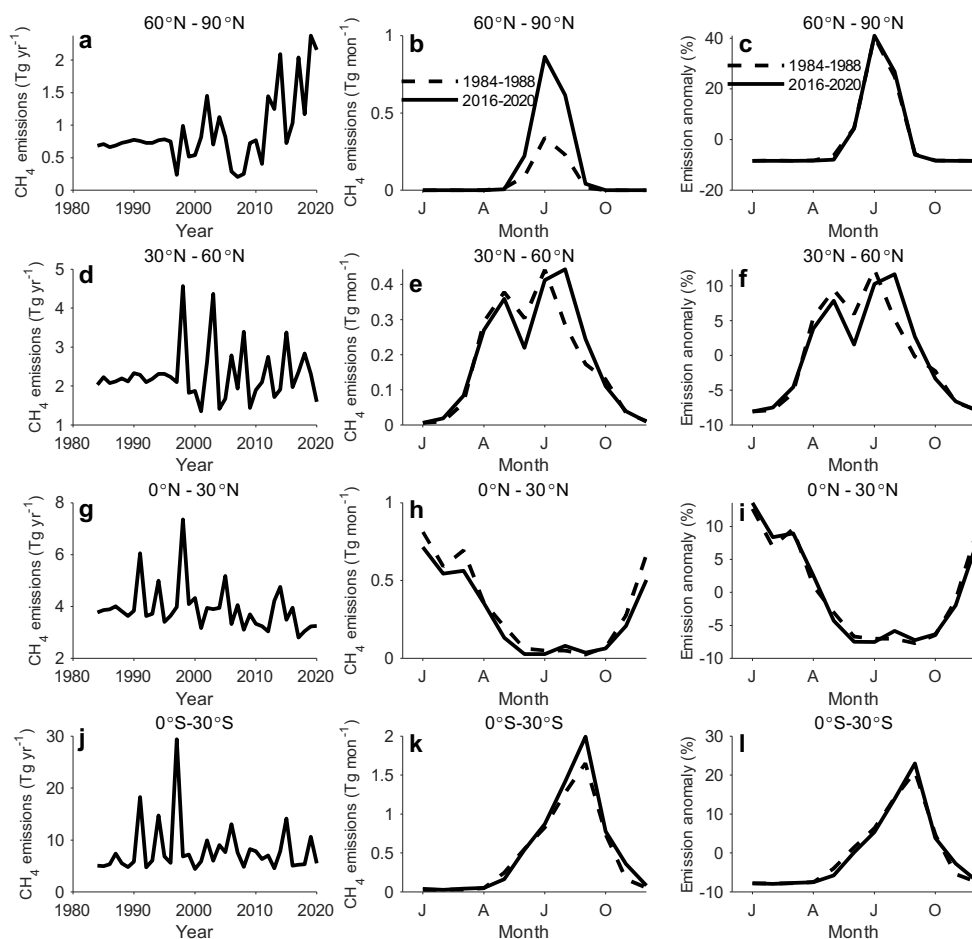
Supplementary Figure 6. Mechanisms and factorial experiments for explaining the changes in seasonal CH₄ amplitude. Increasing emissions amplitude could lead to a lower CH₄ seasonal amplitude, while increasing OH sink amplitude could result in a higher CH₄ seasonal amplitude (**a**). Atmospheric transport could contribute to a decrease of amplitude via more summer CH₄ mass flux from high latitudes, and vice versa. We designed three transportation experiments to quantify the contributions of emissions, sinks and atmospheric transport to the trends in SCA (**b**): T1 experiment with varying emissions (E_{vary}) and sink (S_{vary}); T2 experiment with emissions (E_{1984}) and sink (S_{1984}) held constant at their 1984 levels; T3 experiment with varying emissions but sink held at its 1984 level. Thus, the contribution of the sink can be estimated as the SCA in T1 minus that in T3 and the contribution of emissions can be estimated as the SCA in T3 minus that in T2. The symbols used in this figure are from Integration and Application Network (ian.umces.edu/media-library) under Attribution-ShareAlike 4.0 International (CC BY-SA 4.0).



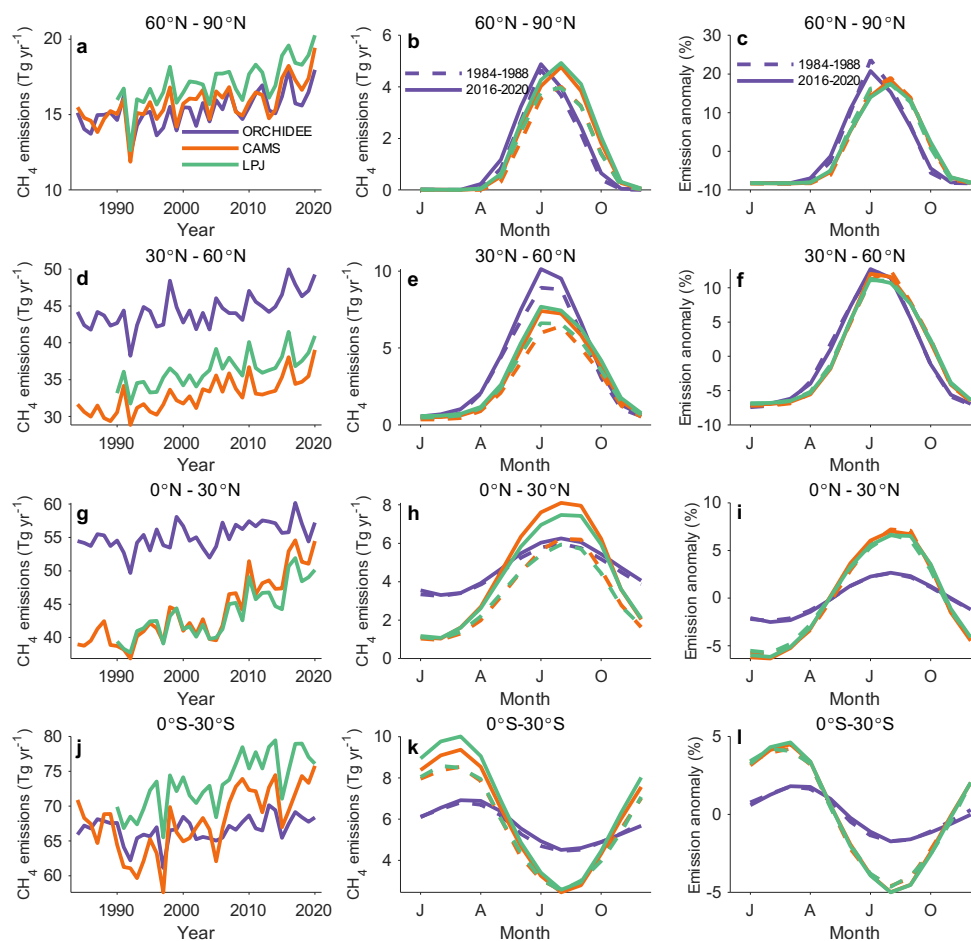
Supplementary Figure 7. Changes of anthropogenic CH₄ emissions between 1984 and 2020 from EDGAR v7.0 and CEDS-2021 for the four latitudinal bands. Annual anthropogenic emissions include the inventories from EDGAR v7.0, CEDS-2021 and a joint product created for this study (see Methods). The changes of absolute monthly anthropogenic emissions (b, e, h, k**), and relative seasonal magnitudes of emissions (**c, f, i, l**) of this study for the four latitudinal bands are shown.**



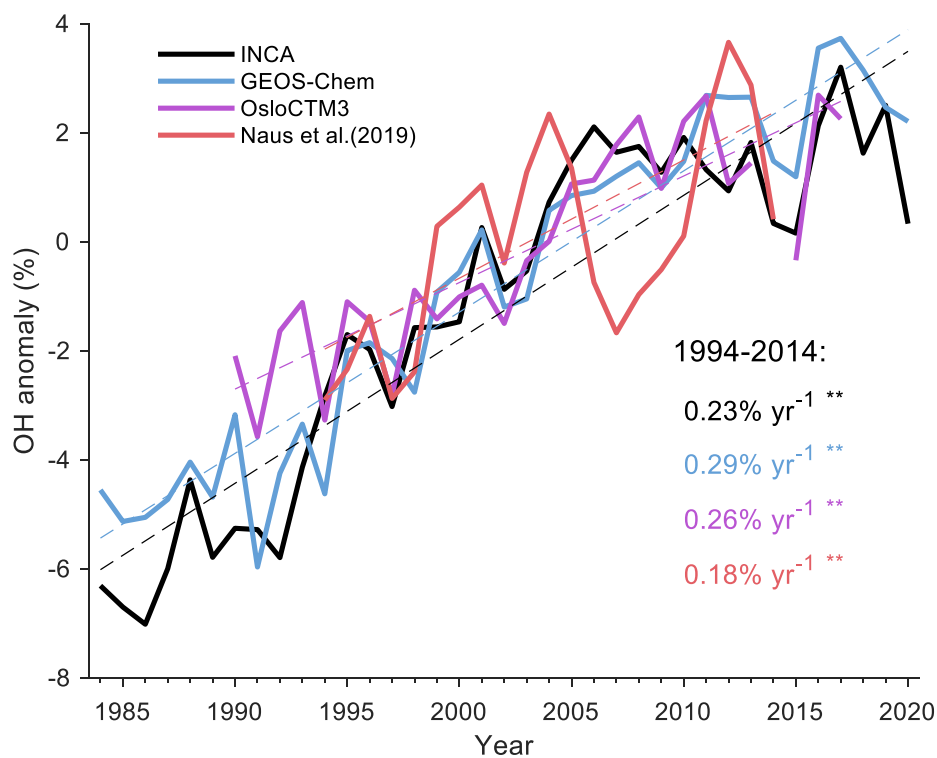
Supplementary Figure 8. Changes of annual and seasonal biomass burning CH₄ emissions between 1984 and 2020 from GFED v4.1s for the four latitudinal bands. The changes of absolute monthly fire emissions (b, e, h, k). and relative seasonal magnitudes of emissions (c, f, i, l) of this study for the four latitudinal bands are shown.



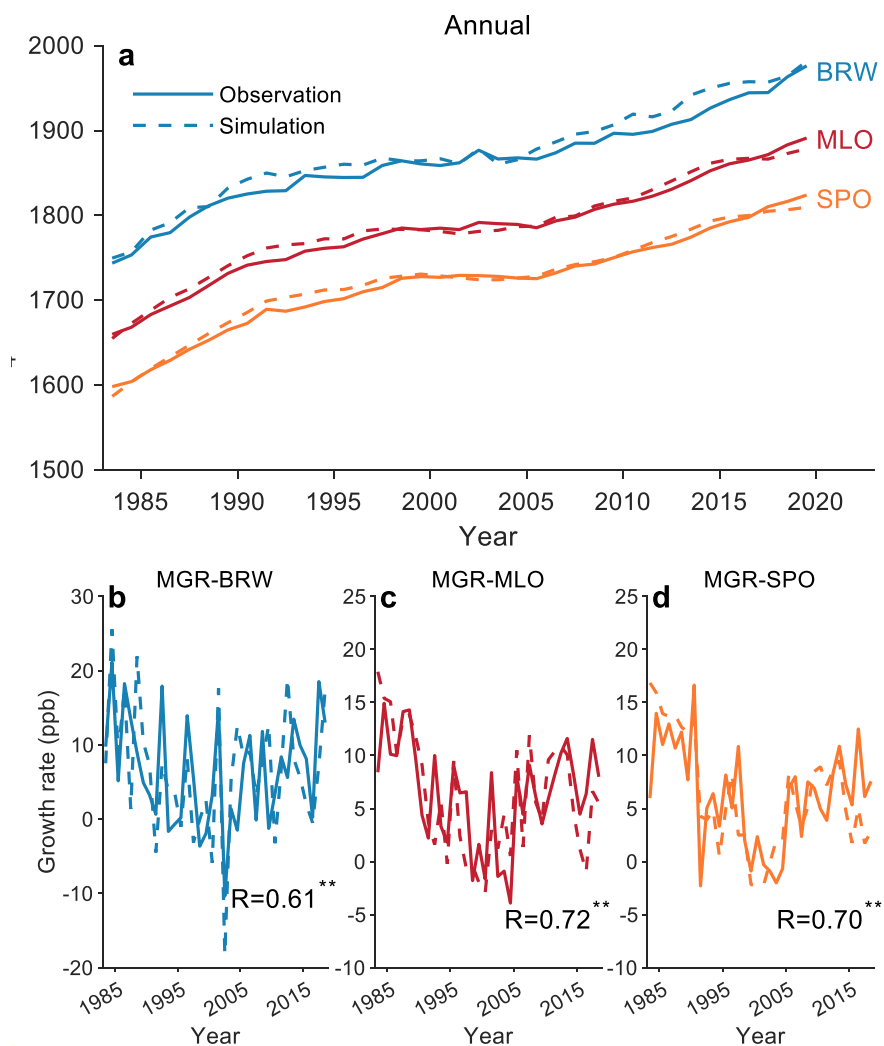
Supplementary Figure 9. Changes of annual and seasonal wetland CH₄ emissions between 1984 and 2020 from ORCHIDEE, LPJ and CAMS for the four latitudinal bands. The changes of absolute monthly fire emissions (b, e, h, k). and relative seasonal magnitudes of emissions (c, f, i, l) of this study for the four latitudinal bands are shown.



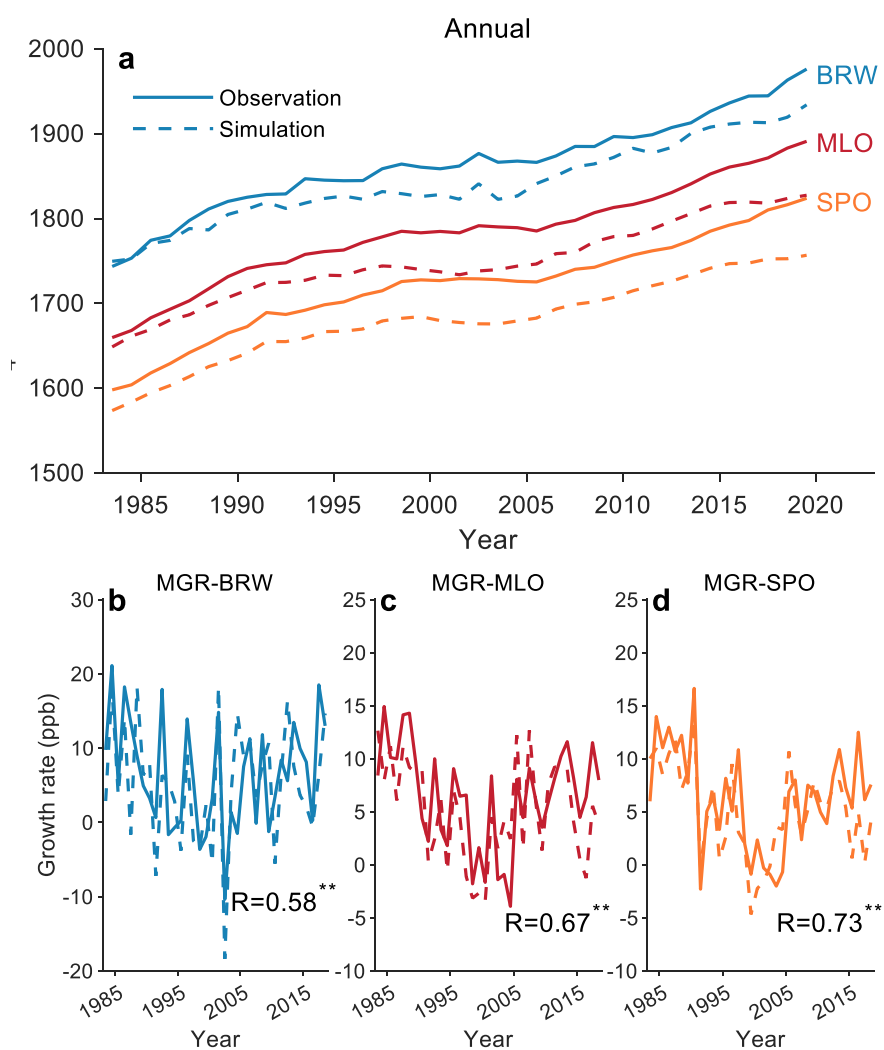
Supplementary Figure 10. Anomalies in global OH concentration simulated by process-based chemistry models INCA, GEOS-Chem and OsloCTM3 (ref. ²⁷), and derived from MCF by Naus et al. (ref. ²⁸). The regression coefficients of the global OH concentration in 1994-2014 are also marked in the lower right of each panel, with ** for $p < 0.01$.



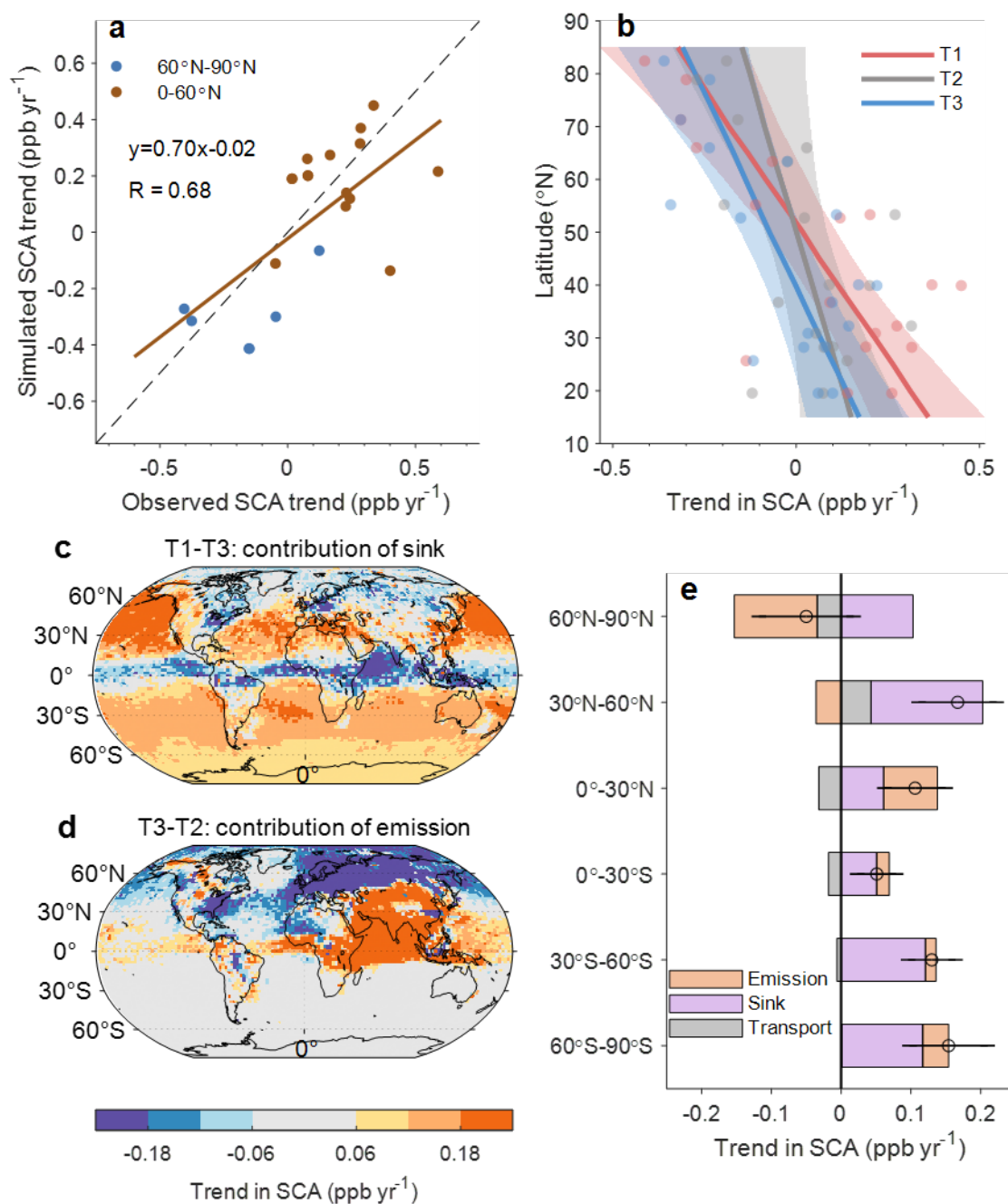
Supplementary Figure 11. Simulations of CH₄ concentrations between 1984 and 2020 from GEOS-Chem forced by OH concentration from INCA. The performance of GEOS-Chem on simulating the annual concentration **(a)** and methane growth rate at Barrow **(b)**, Mauna Loa **(c)** and South Pole **(d)**. The correlation coefficients between the observed methane growth rate and simulations are also marked in the lower right of each panel, with ** for $p < 0.01$.



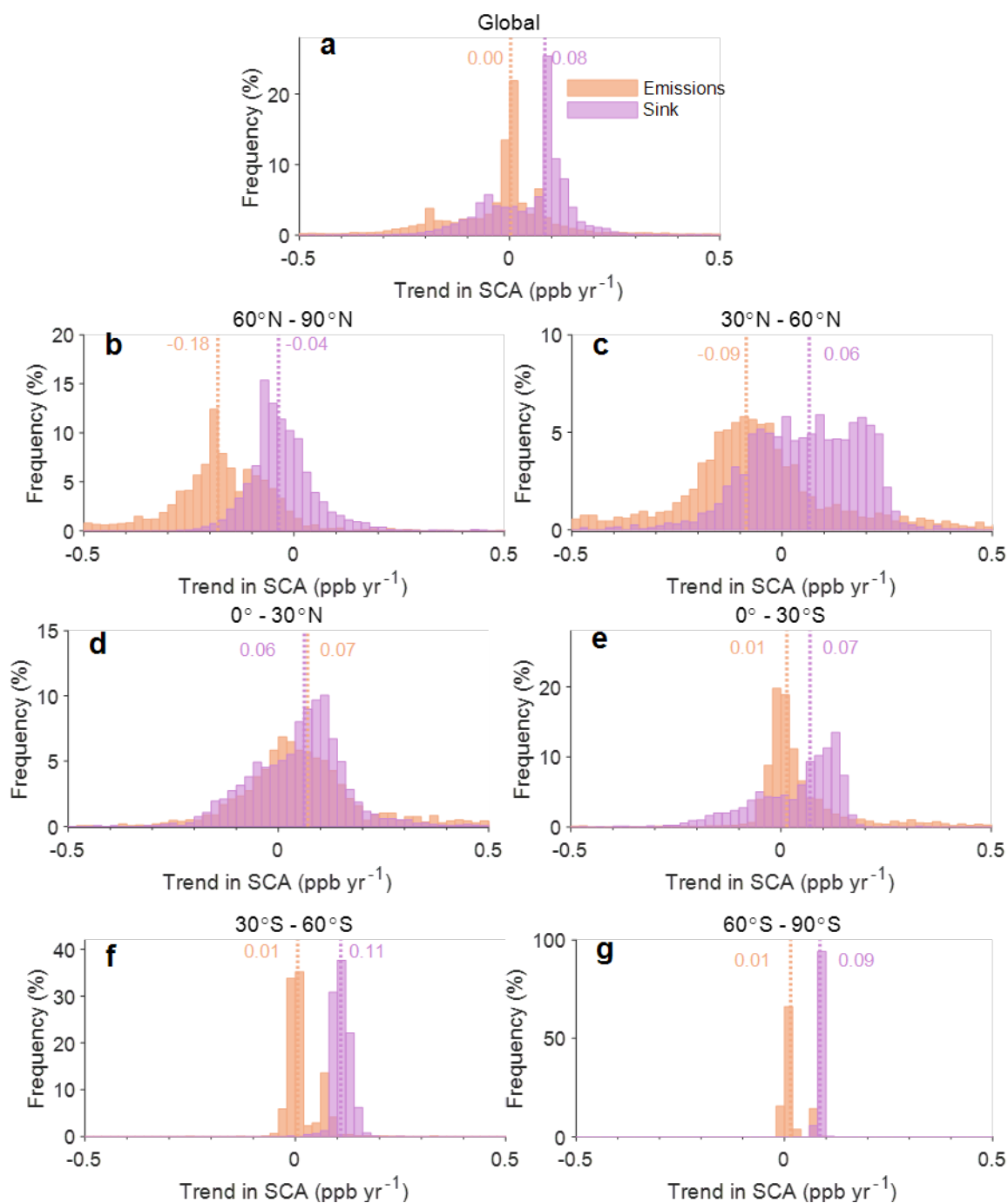
Supplementary Figure 12. Simulations of CH₄ concentrations between 1984 and 2020 from GEOS-Chem forced by OH concentration from GEOS-Chem. The performance of GEOS-Chem on simulating the annual concentration (**a**) and methane growth rate at Barrow (**b**), Mauna Loa (**c**) and South Pole (**d**). The correlation coefficients between the observed methane growth rate and simulations are also marked in the lower right of each panel, with ** for $p < 0.01$.



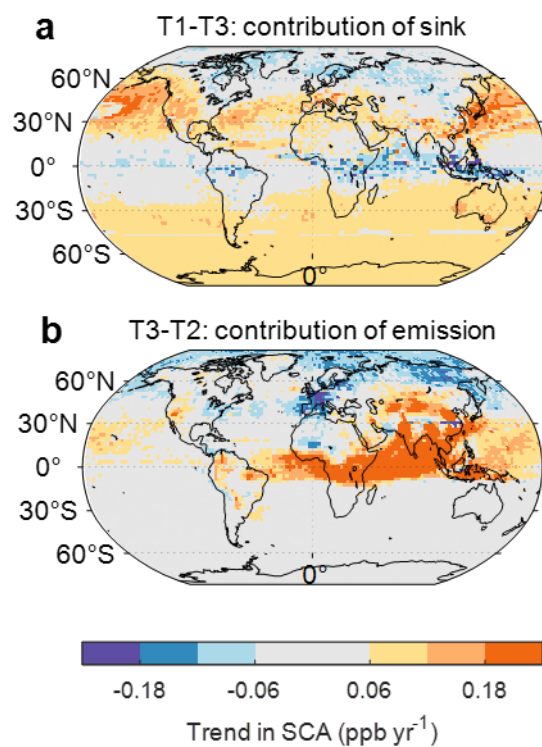
Supplementary Figure 13. Same as Figure 2 in the main text, except that GEOS-Chem OH concentrations, rather than those from INCA, were used to force GEOS-Chem simulations.



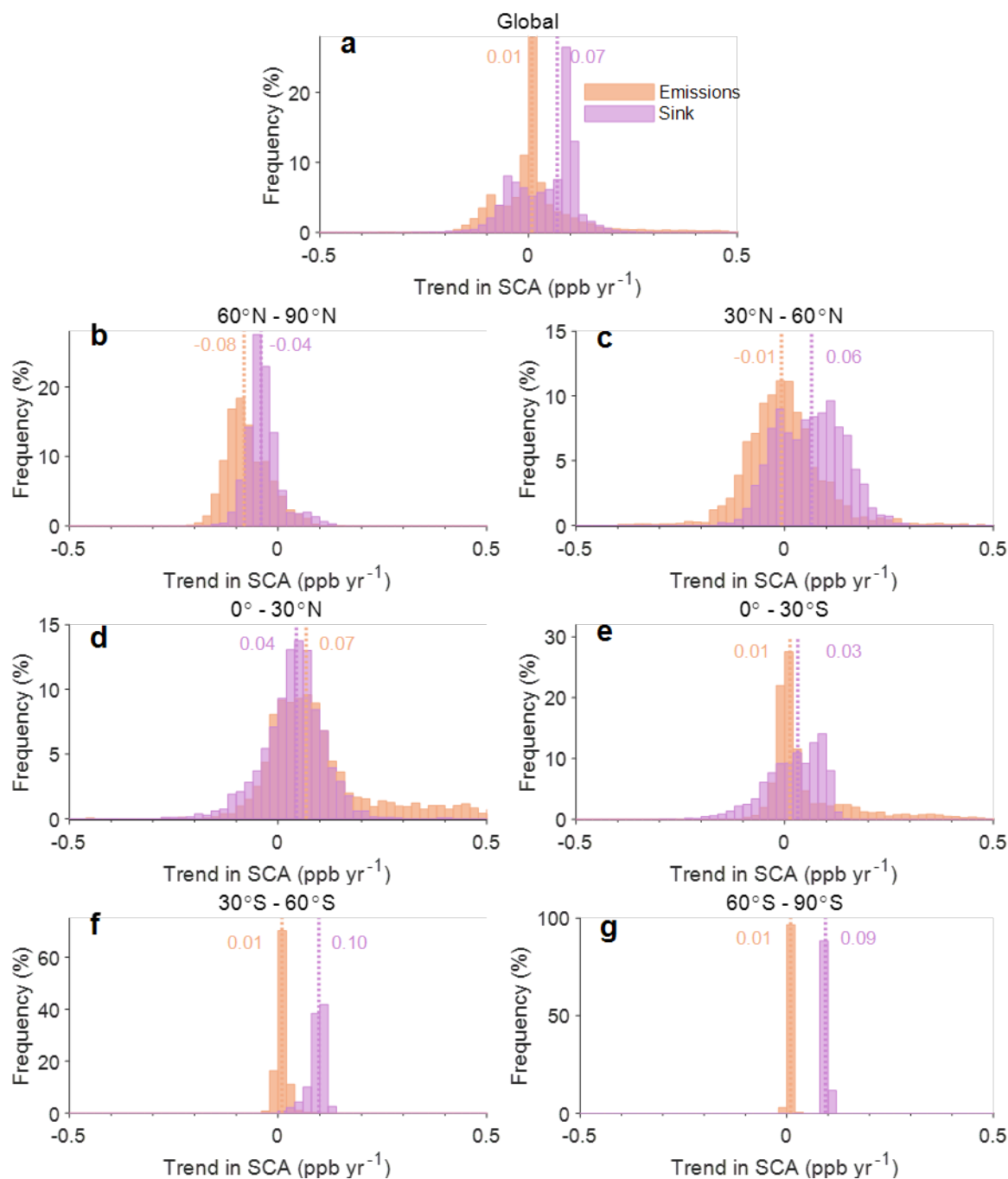
Supplementary Figure 14. Histograms of the distributions of contribution of emission and sink to the trend in surface-level CH₄ SCA between 1984 and 2020 across all pixels for the entire globe (a) and six latitudinal bands (b-g). The dashed lines represent the medians.



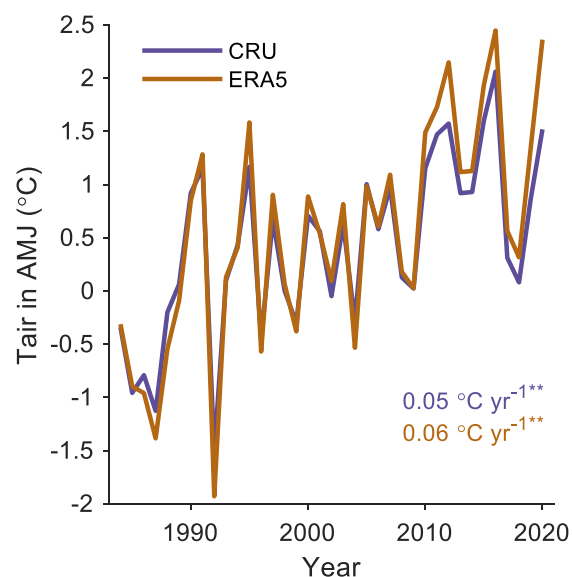
Supplementary Figure 15. The spatial patterns of the contributions of sink and emission to the trend in tropospheric CH₄ SCA. Same as Figure 2c and d but for the tropospheric CH₄ SCA.



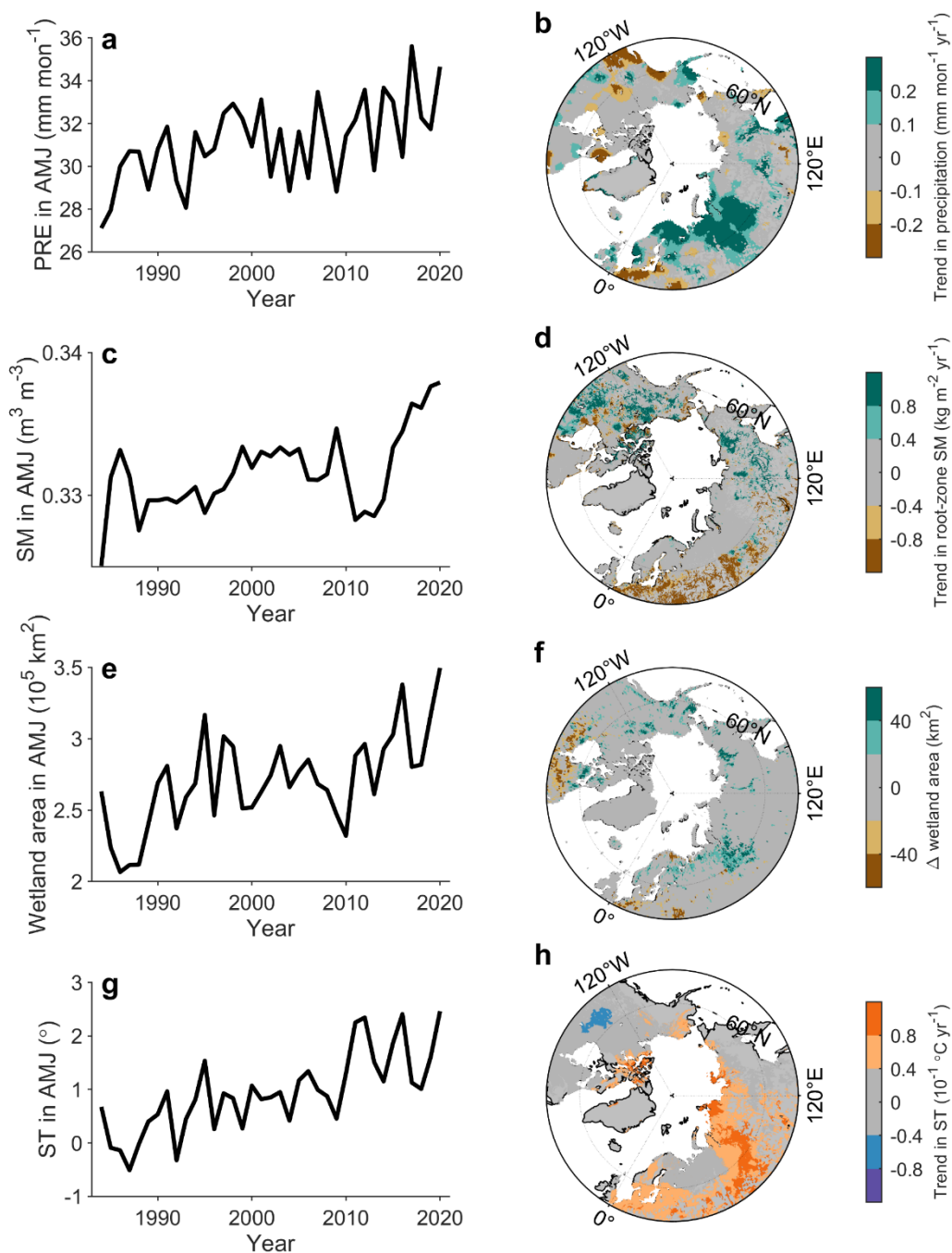
Supplementary Figure 16. Histograms of the distributions of the contribution of emission and sink to the trend in tropospheric CH₄ SCA between 1984 and 2020 across all pixels of the entire globe (a) and six latitudinal bands (b-g). The dashed lines represent the medians.



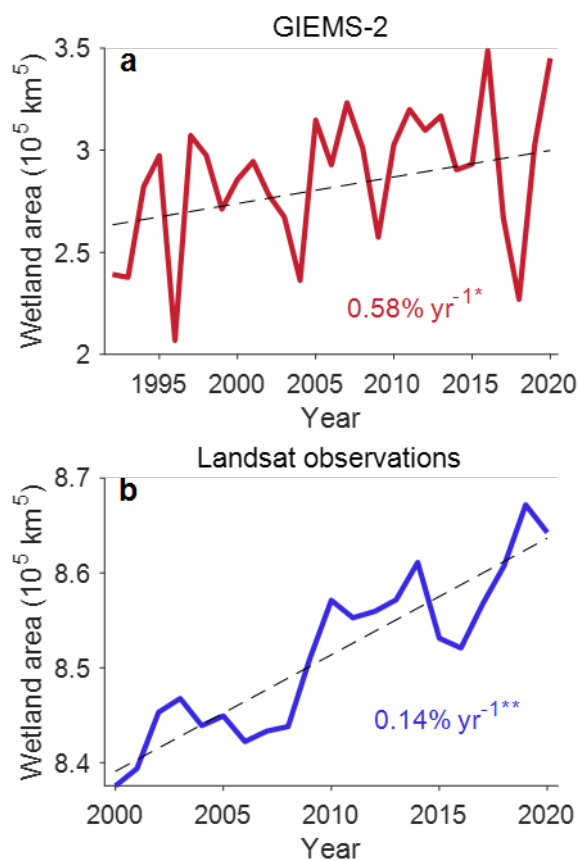
Supplementary Figure 17. The changes in surface air temperature over northern high-latitude wetlands. The surface air temperature over northern high-latitude wetlands in April, May and June (AMJ) from CRU²⁹ and ERA5 (ref. ³⁰). The regression coefficients of the surface air temperature in 1984- 2020 are also marked in the lower right of each panel, with ** for $p < 0.01$.



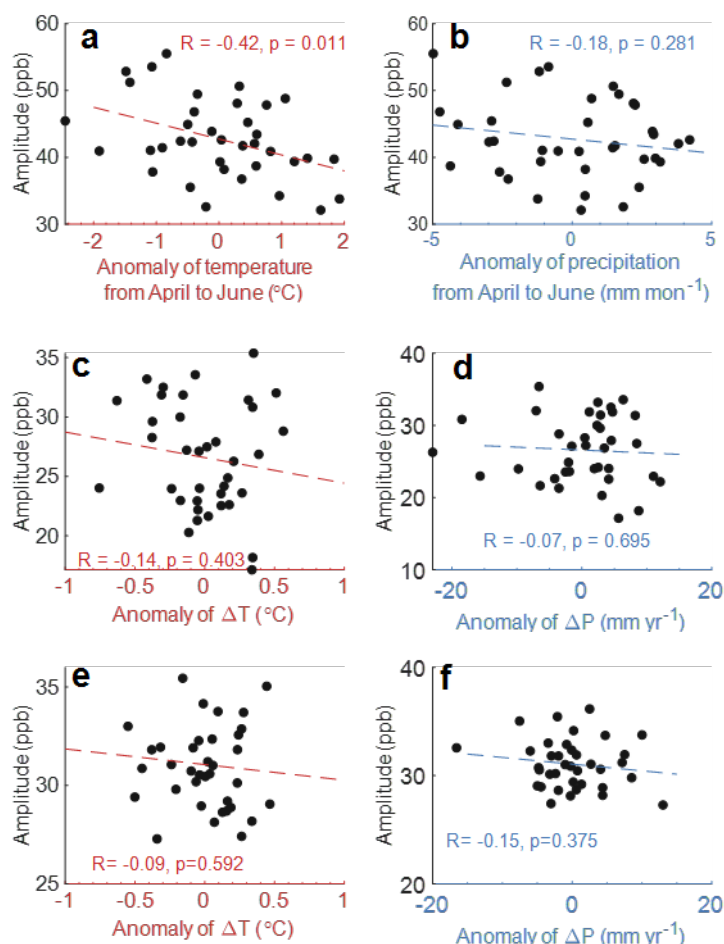
Supplementary Figure 18. The changes in precipitation, soil moisture, wetland area and soil temperature over northern high-latitude wetlands. The precipitation from CRU²⁹ (a), soil moisture from GLEAM v3.8a (c; ref. ³¹), wetland area simulated by TOPMODEL^{32,33} (e) and soil temperature from ERA5 (g; ref. ³⁰) over northern high-latitude wetlands in April, May and June (AMJ). The spatial patterns of the trend in precipitation (b), soil moisture (d), soil temperature (h) and changes in wetland area (f) in Arctic regions are also shown.



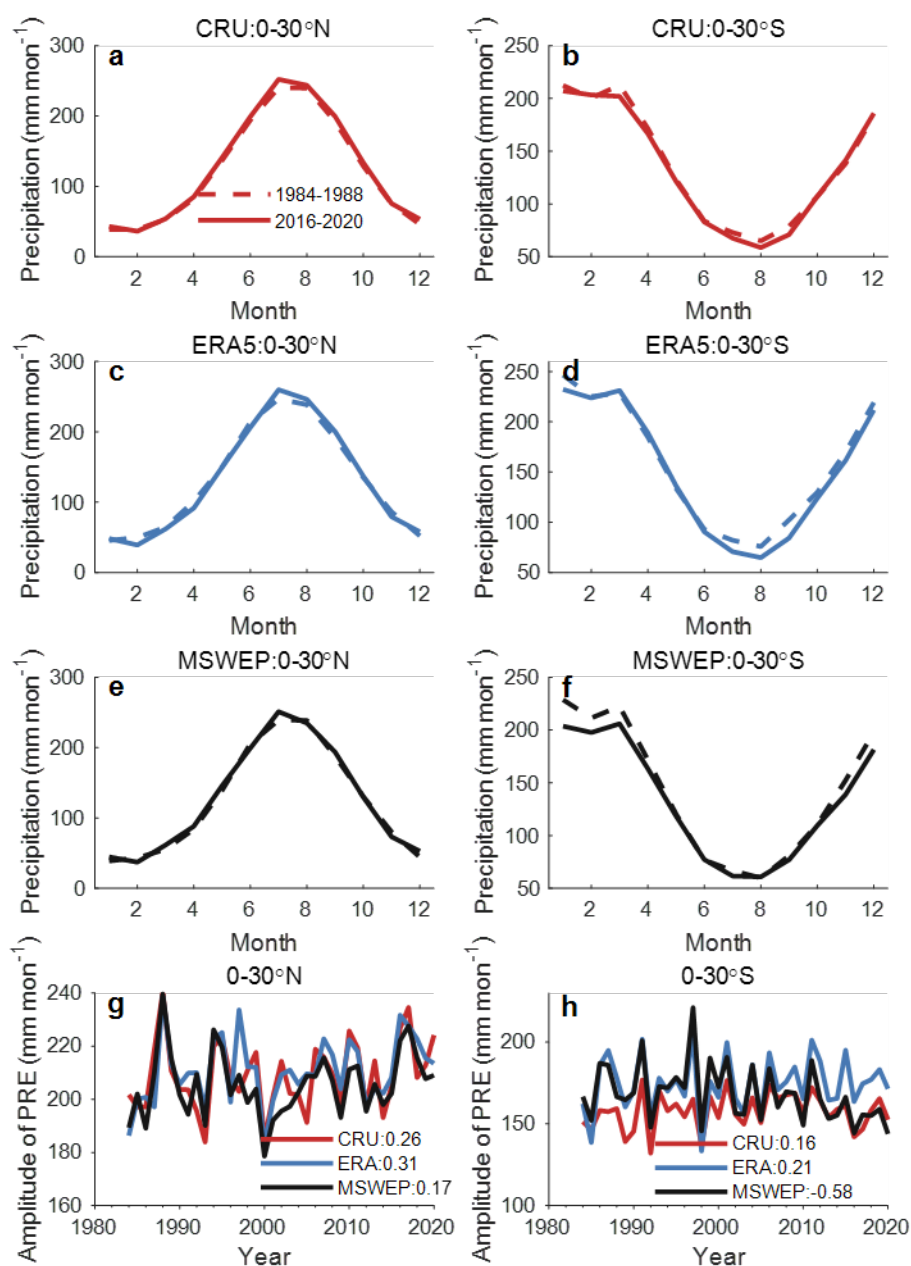
Supplementary Figure 19. Changes of wetland area over northern high latitudes from two satellite-based observations. The wetland area in April, May and June (AMJ) over northern high latitudes from GIEMS-2 between 1992-2020 (**a**; ref. ³⁴) and annual wetland annual area between 2000-2020 from Landsat observations (**b**; ref. ³⁵).



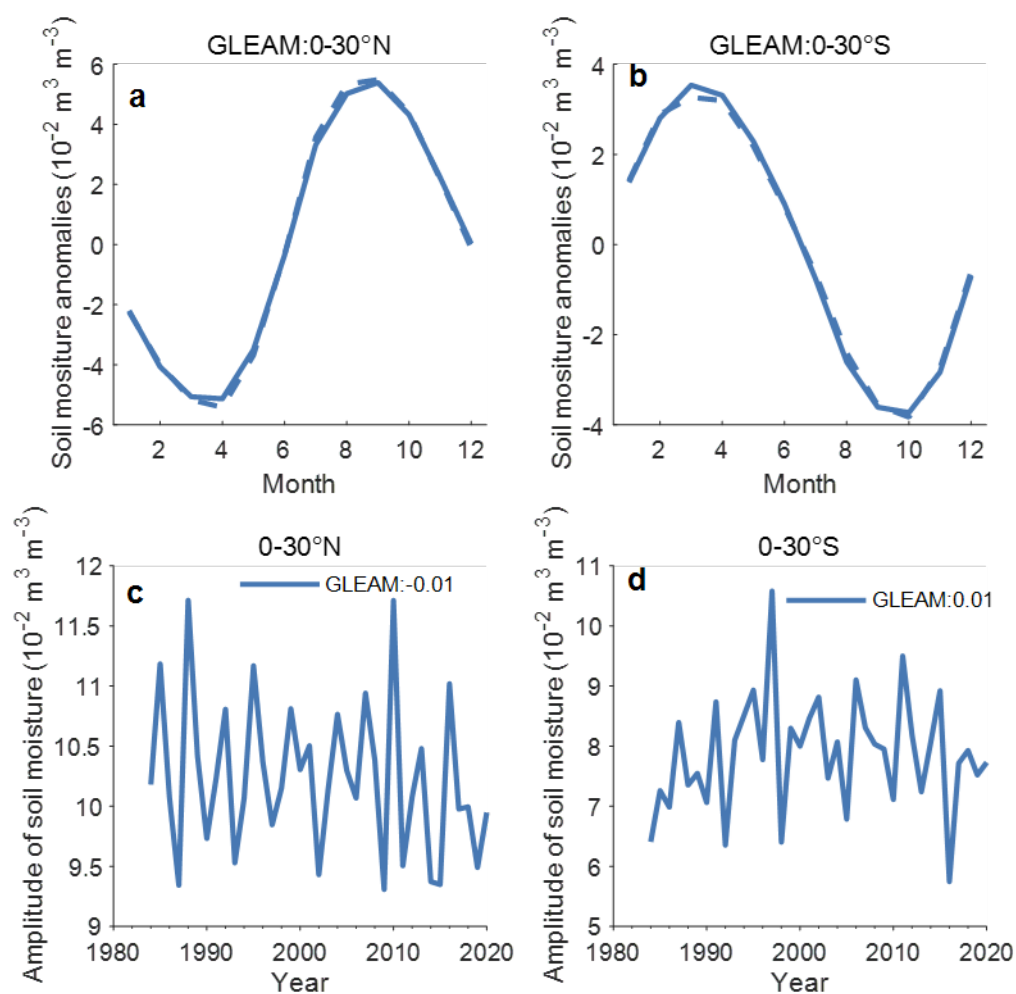
Supplementary Figure 20. Temperature and precipitation anomalies against the observed amplitude between 1984 and 2020. Anomaly of temperature and precipitation from April to June over high-latitude northern wetlands in 1984-2020 from ERA5 against the observed CH_4 amplitude at BRW (**a**, **b**). Amplitude of temperature and precipitation (defined as average temperature/precipitation in May to July minus February to April) over northern tropical wetlands from ERA5 against the observed SCA at MLO (**c**, **d**) and over southern tropical wetlands against the observed SCA at SPO (**e**, **f**).



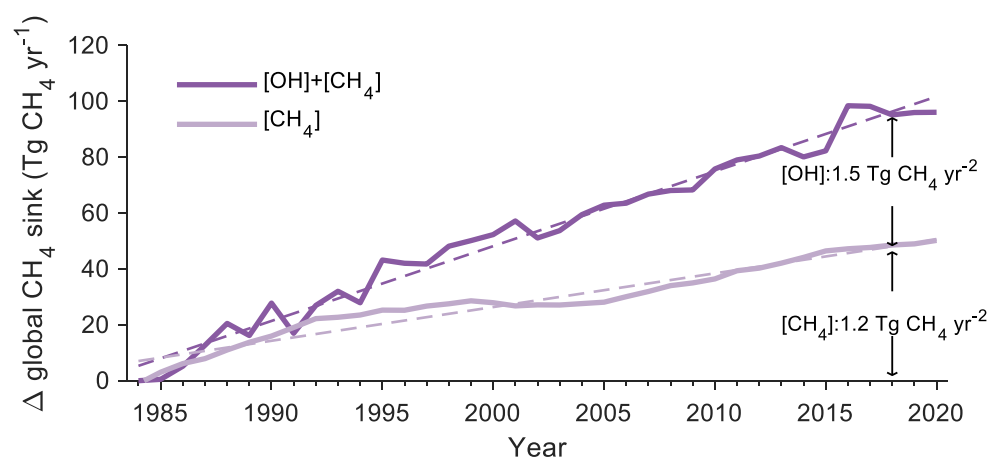
Supplementary Figure 21. Changes of seasonal cycle of tropical wetland precipitation. The seasonal cycle of northern tropical wetland precipitation during 1984-1988 and 2016-2020 from CRU (a), ERA5 (c) MSWEP (e), and changes in their seasonal amplitude between 1984 and 2020 (g). The seasonal cycle of southern tropical wetland precipitation is also shown for the same periods and data sources (b, d, f, h).



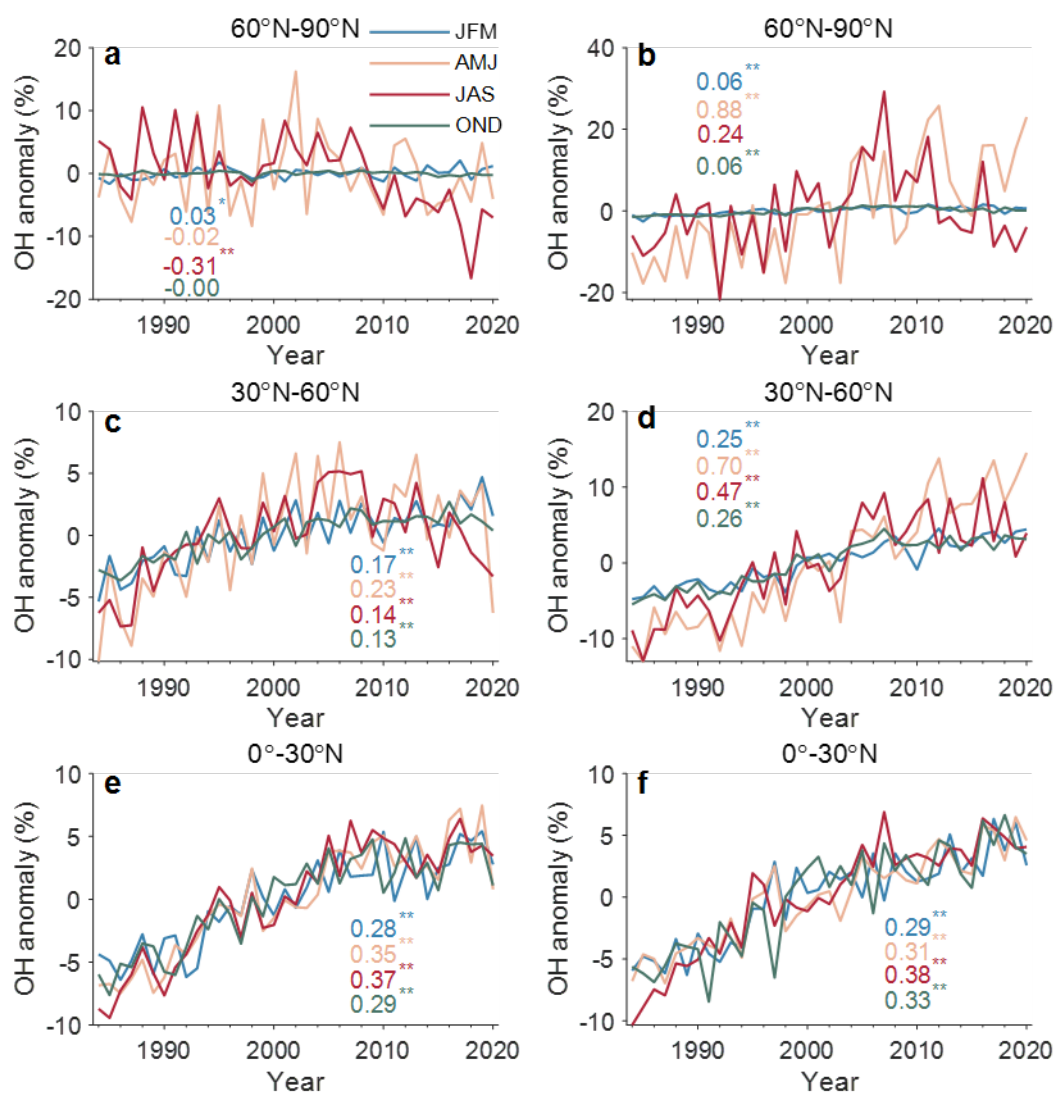
Supplementary Figure 22. Changes in the seasonal cycle of tropical wetland soil moisture between 1984 and 2020 from GLEAM.



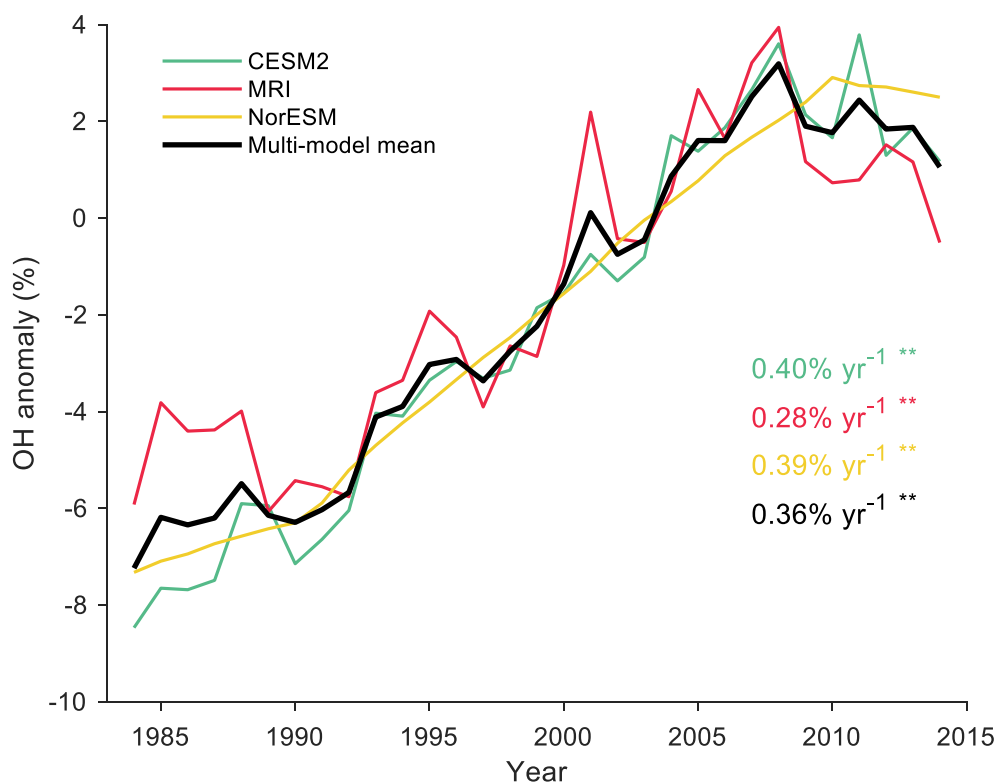
Supplementary Figure 23. Same as Figure 4a in the main manuscript, except that GEOS-Chem OH concentrations, rather than those from INCA, were used in the GEOS-Chem simulation.



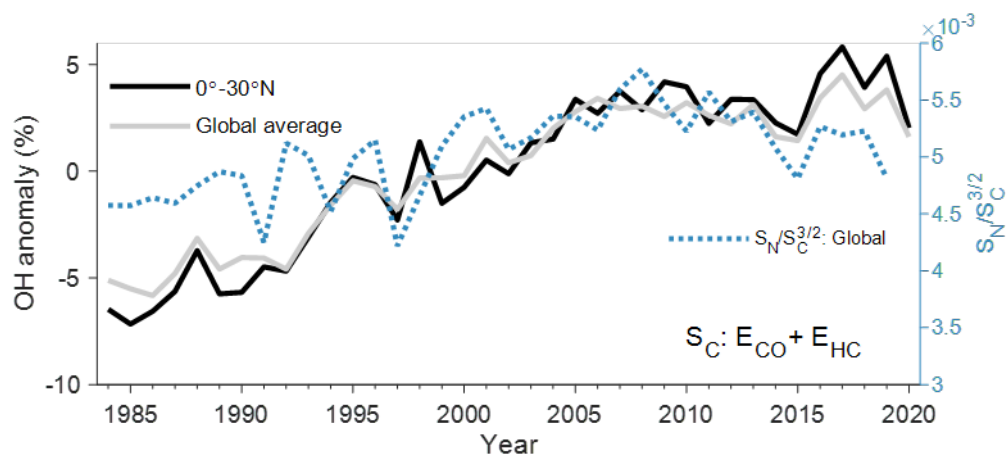
Supplementary Figure 24. Seasonal OH concentration anomalies in three latitudinal bands from INCA (a, c, e) and GEOS-Chem (b, d, f).



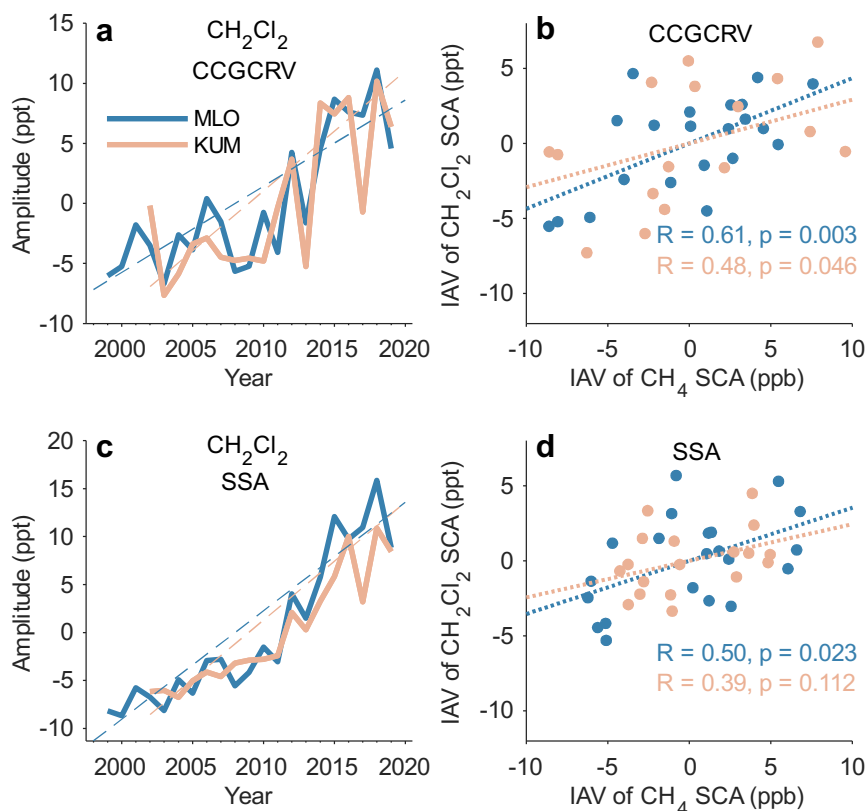
Supplementary Figure 25. Anomalies of global OH concentration simulated by ESMs. The OH concentration anomalies simulated by CESM2, NorESM, MRI and the Multi-model mean (black line) are shown. The regression coefficients of the global OH concentration in 1984-2014 are also marked in the lower right of each panel, with ** for $p < 0.01$.



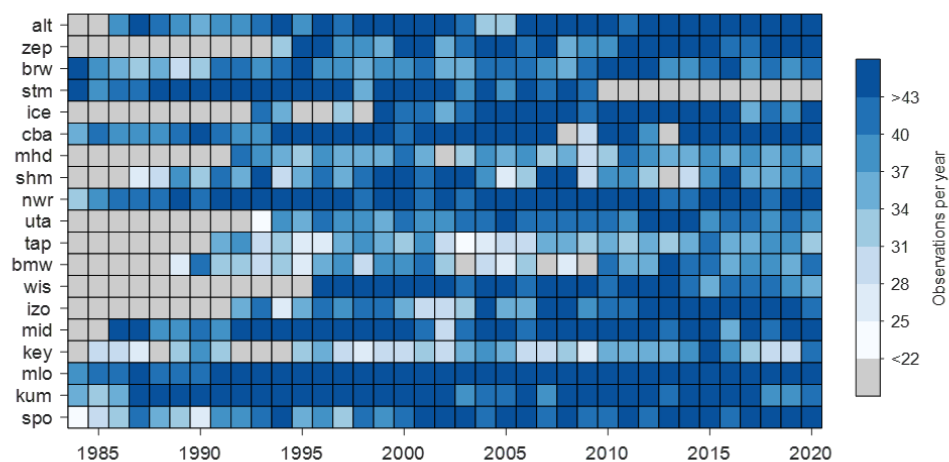
Supplementary Figure 26. Same as Figure 4b in the main manuscript, except with the simple proxy index $S_N/S_C^{3/2}$ for tropospheric OH concentration. S_N and S_C are the sources of NO_x and of CO and hydrocarbons, respectively. The anthropogenic sources of hydrocarbons are from CEDS-2021 and biomass burning sources from GFED v4.1s.



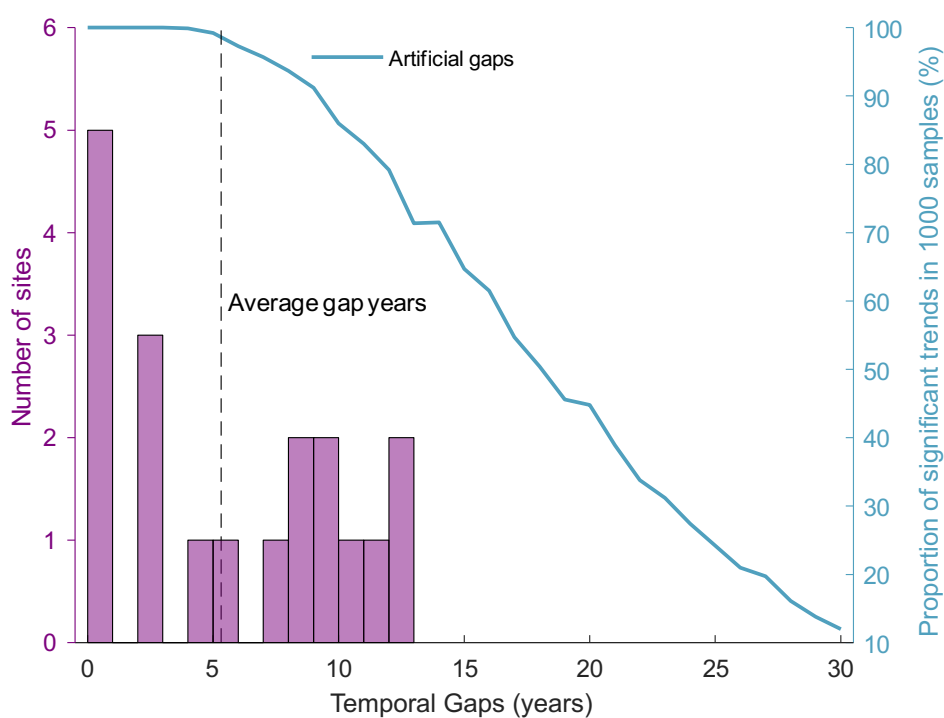
Supplementary Figure 27. The observed SCA of CH_2Cl_2 and the correlations between interannual variability (IAV) of CH_4 SCA and CH_2Cl_2 SCA. The observed SCA of CH_2Cl_2 at MLO and KUM extracted by CCGCRV (a) and SSA (c). The correlations between interannual variability (IAV) of CH_4 SCA and CH_2Cl_2 SCA at both MLO and KUM extracted by CCGCRV (b) and SSA(d). The correlation coefficients between the CH_4 SCA and CH_2Cl_2 SCA are also marked in the lower right of each panel.



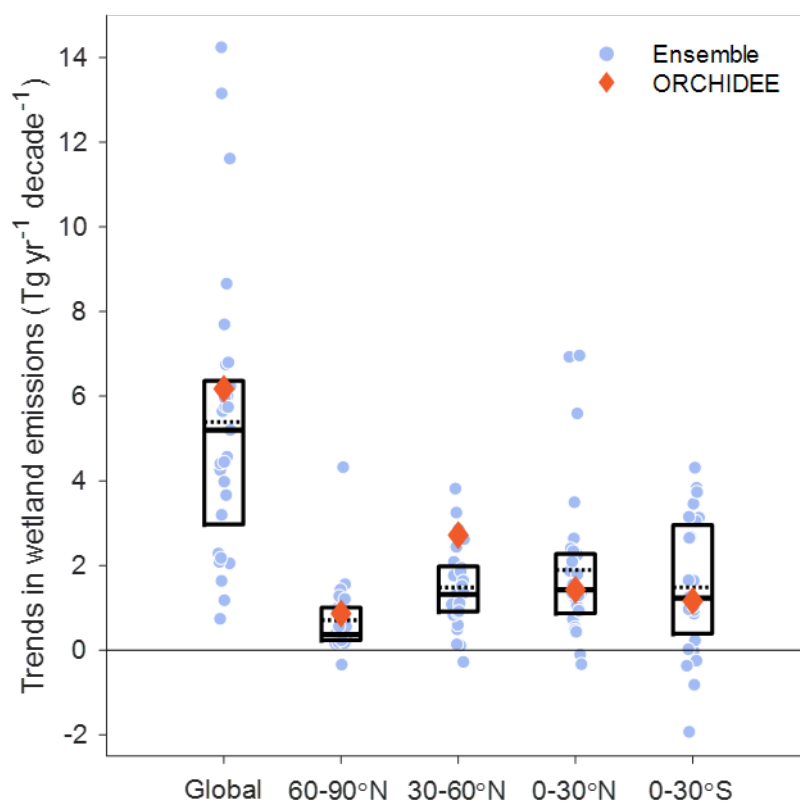
Supplementary Figure 28. The number of 8-days intervals when there was at least one observation for all sites from 1984 to 2020 (without missing data, there must be 46 8-day flasks in a year). The grey squares indicate observations fewer than 22 times in these years, which were excluded in the SCA trend analysis.



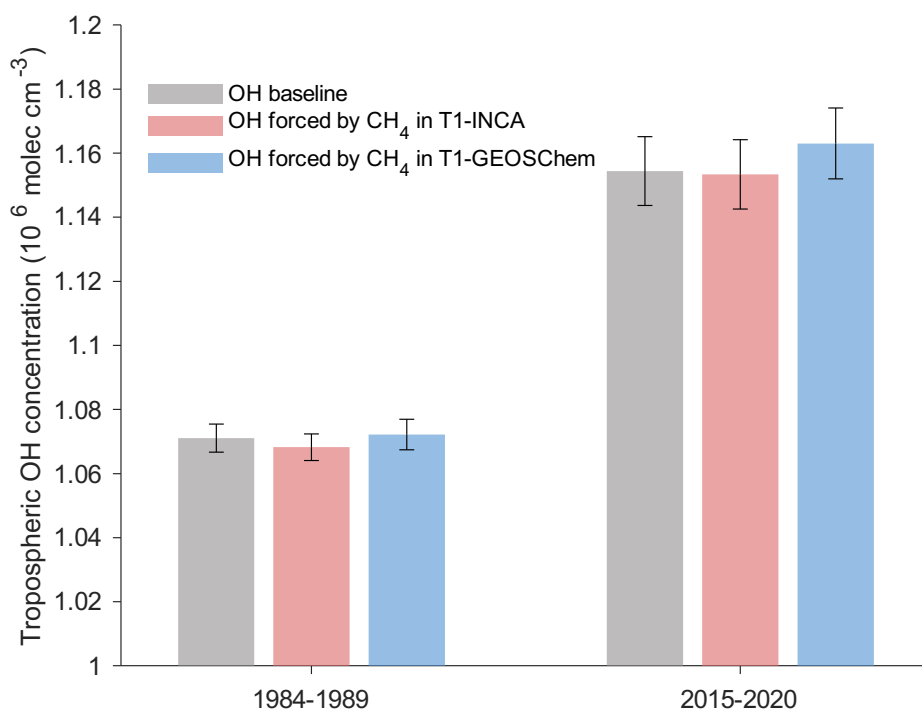
Supplementary Figure 29. The distribution of temporal gap years from 19 sites and the test for the proportion of 1000 Monte Carlo samples with significant trends changes with artificial gap years at MLO. The 1000 time series samples with artificial gap years, ranging from 1 year to 30 years, are selected randomly by Monte Carlo method. These artificial gap years are excluded in calculating the SCA trend at MLO, and the proportion of the samples with significant SCA trends ($p < 0.05$, as the “true” significant SCA trend detected with full 37 years time series at MLO, i.e., no missing year) in the 1000 samples along with the number of artificial gap years.



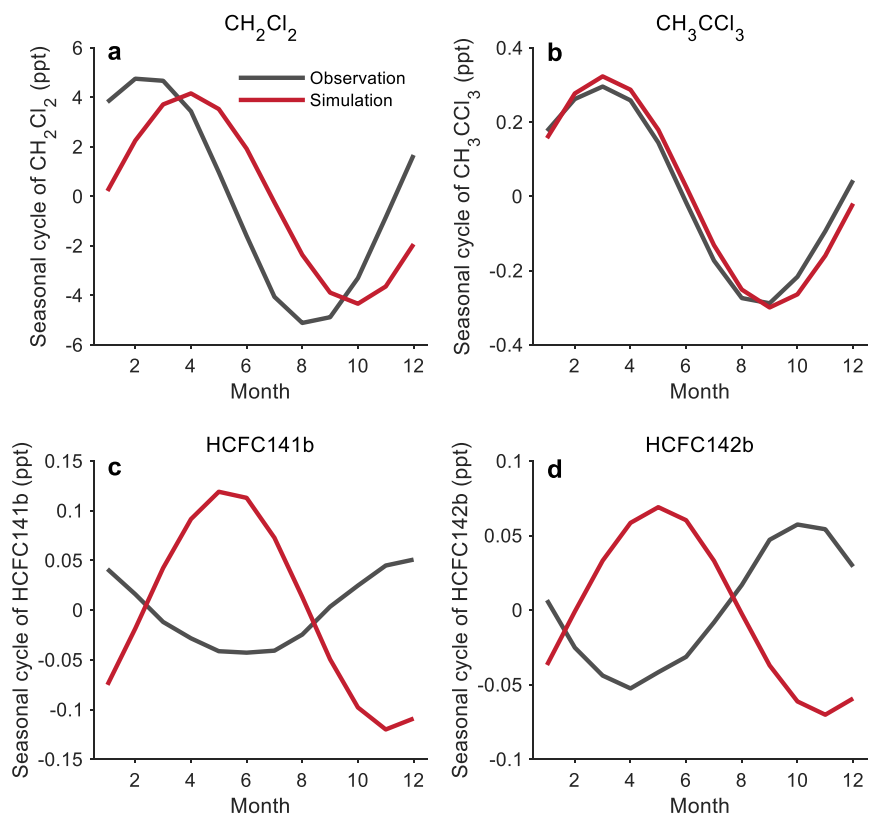
Supplementary Figure 30. The trends in global wetland emissions and for four latitude bands during the period 2000-2020, from the 31 available datasets. The 31 datasets of wetland emissions are from modeling results of ORCHIDEE, LPJ-EOSIM¹ and the Global Carbon Project^{2,3} (GCP-CH₄), WETCHARTS⁴, and the atmospheric inversion from CAMS⁵, as well as upscaling wetland methane emissions (UpCH₄)⁶ and Boreal-Arctic wetland emissions from the causality-guided machine learning model⁷. The lower, middle and upper solid lines in the box plot represent the 25th percentile, the median and the 75th percentile, while the dashed line represents the mean.



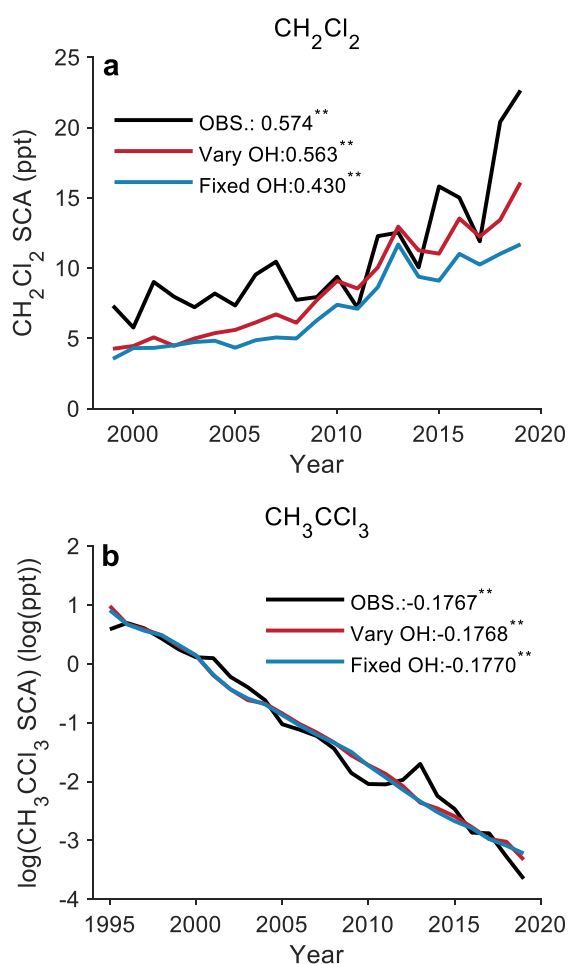
Supplementary Figure 31. Global average of tropospheric OH concentration forced by atmospheric CH₄ concentration from observations and T1 experiments. Average tropospheric GEOS-Chem OH concentration forced by CH₄ concentration from observations (OH baseline, the grey bars) during 1984-1989 and 2015-2020. The tropospheric OH concentration forced by CH₄ concentration in T1 experiment simulated by OH concentration from INCA (OH forced by CH₄ in T1-INCA, the red bars) and GEOS-Chem (OH forced by CH₄ in T1-GEOSChem, the blue bars) during 1984-1989 and 2015-2020. The error bars represent the standard deviation of the OH concentration of 1984-1989 and 2015-2020.



Supplementary Figure 32. Seasonal variations of CH_2Cl_2 , CH_3CCl_3 , HCFC-141b and HCFC-142b observed at MLO and simulated by MALTA.



Supplementary Figure 33. Long-term trends in SCA of CH_2Cl_2 and CH_3CCl_3 from observations at MLO and from MALTA M1 simulation with varying OH concentration and T2 simulation with fixed OH concentration. The trends in SCA of CH_2Cl_2 from observations and simulations are marked in the upper of each panel, with ** for $p < 0.01$.



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